

Eastern and South-Eastern Asia Research and Action Agenda

List of Authors

Renzo R. Guinto – Planetary and Global Health Program, St. Luke's Medical Center College of Medicine - William H. Quasha Memorial, Philippines

John Jamir Benzon R. Aruta – Planetary and Global Health Program, St. Luke's Medical Center College of Medicine and Department of Psychology, De La Salle University, Philippines

Gail S. Co – Planetary and Global Health Program, St. Luke's Medical Center College of Medicine - William H. Quasha Memorial, Philippines

Kent Tristan Esteban - Planetary and Global Health Program, St. Luke's Medical Center College of Medicine - William H. Quasha Memorial, Philippines

Dianne J. Cunanan – Planetary and Global Health Program, St. Luke's Medical Center College of Medicine - William H. Quasha Memorial, Philippines

Jun Angelo Sunglao - Planetary and Global Health Program, St. Luke's Medical Center College of Medicine - William H. Quasha Memorial, Philippines

Alfredo Lorenzo R. Sablay – Planetary and Global Health Program, St. Luke's Medical Center College of Medicine - William H. Quasha Memorial, Philippines

Joji Sugawara - Health and Global Policy Institute, Japan

Niaya Harper Igarashi – Health and Global Policy Institute, Japan

Table of contents

Executive summary	3
Introduction	5
Background to Connecting Climate Minds	6
Methodology	8
Current state and emerging needs for climate change and mental health in Eastern and South-Eastern Asia	14
Research agenda	24
Priority research themes	24
Action agenda	31
Discussion: strengths, limitations and next steps for the research and action agenda	36
Conclusion	37
Hearing from the regional community of practice	38
Glossary of terms	39
Who produced this report	39
Appendix	44
References	64

Executive summary

Climate change is increasingly recognised as a threat to mental health, compounding risks for poor mental health outcomes and destabilising the conditions needed for good mental health. In Eastern and South-Eastern Asia (ESEA), abrupt climate-related events such as typhoons, storm surges and landslides, as well as slow-onset events such as sea-level rise, increasing temperature and drought, continue to put individuals at risk of both physical and psychological harm.

While research at the intersection of climate change and mental health has proliferated in recent years, the field remains disconnected, unequal and siloed, slowing urgent progress to address the mental health impacts of climate change and gain the mental health benefits of climate action. In ESEA, research around this important nexus is scarce and a disconnect remains between the limited findings and actual policy, highlighting the need for more research to bridge this gap and inform decision-making.

Connecting Climate Minds (CCM) is a Wellcome-funded initiative to cultivate a collaborative, transdisciplinary climate change and mental health field with a clear and aligned vision. Over the last year, we have convened experts across disciplines, sectors, and countries to develop regional and global research and action agendas. These agendas set out 1) research priorities to understand and address the needs of people experiencing the mental health burden of the climate crisis, and 2) priorities to enable this research and translate evidence into action in policy and practice. We hope this work will help to focus the efforts of funding, policy, and practice communities working toward a future where no one is held back by climate-related mental health challenges.

This report presents the research and action agenda for climate change and mental health in ESEA. A total of 30 priority research themes emerged across the four key research categories of the project:

- **Impacts, risk, and vulnerable groups:** These themes explore: which mental health impacts (e.g., anxiety, depression, trauma and stress) arise from climate change events common in ESEA, including extreme typhoons, flooding, increasing heat, bushfire, drought, etc.; how they manifest; and who is disproportionately affected (e.g., farmers, fisher peoples, low-income communities, children and youth, the elderly, people with disabilities, those with existing health and mental health problems, etc.).
- **Pathways and mechanisms:** These themes tackle how mental health outcomes driven by climate change arise as well as the proximal, intermediate and distal factors that explain the link between climate change and mental health (e.g., socioeconomic and health inequities, gender-based inequalities and violence, stigma, disrupted spiritual connection to land, etc.).
- **Mental health benefits of climate action:** These themes explore how climate mitigation and adaptation actions generate positive mental health outcomes (e.g., improved mental health through individual and community engagement in climate action, sustainable urbanisation, creation of climate-resilient and sustainable healthcare systems, integration of mental health and climate change policies, etc.).
- **Mental health interventions in the context of climate change:** These themes analyse existing and novel mental health prevention and curative approaches applicable to climate change-affected situations (e.g., preventive solutions for mental health caused by abrupt and slow-onset climate events, strengthening mental health and psychological support for climate-related disasters,

community-based and culturally sensitive climate-mental health interventions, climate change and mental health education as an intervention).

Our regional vision is an ESEA where citizens and communities enjoy the highest attainable level of mental health, living in harmonious balance, resilience, equity and justice, even in the face of a changing climate.

To advance the climate change and mental health agenda in the region, we have identified a number of priority actions. To create an enabling environment for research, this includes: enhanced funding support for climate change and mental health research; engaging people and communities outside academia; and advocating for more climate change and mental health research within academic institutions.

Additionally, it is essential to effectively translate the growing evidence base into action so that the region can better respond to the mental health impacts of climate change. Opportunities to do so include: encouraging meaningful collaboration between researchers from different disciplines, policymakers and affected communities to create evidence-based and culturally appropriate interventions and policies; and leveraging the increasing awareness of the public and government leaders to advocate for better policies and interventions that simultaneously address climate change and promote mental health.

Introduction

Context

Climate change and mental health are two of our greatest global challenges, and awareness of the intersection between mental health and the climate crisis has grown rapidly in recent years.¹ Climate change exacerbates mental health challenges by increasing exposure to extreme heat and the traumas of extreme weather events,² destabilising the conditions needed for good mental health and wellbeing (for example, water and food insecurity, forced migration, polluted air, loss of treasured environments),² disrupting access to healthcare,³ and increasing psychological distress through awareness of climate threats and insufficient climate action.⁴ People living with mental health challenges are also particularly vulnerable to the stressors of the climate crisis, such as increased risk of physical heat stress and death during heatwaves.^{5, 6, 7}

In response to the mounting mental health toll of the climate crisis, research in the climate and mental health field has grown rapidly. Nevertheless, key evidence gaps exist in many regions, including the mental health burden attributable to climate change, the pathways and mechanisms underlying these impacts, the co-benefits of climate action for mental health and the best interventions or solutions to support mental health in a changing climate. Climate change and mental health research remains frustratingly disconnected across disciplines, sectors, and geographies, and is unevenly focused on certain topics and global regions.⁸ Moreover, siloed decision making slows the translation of evidence to aligned action across climate and mental health policy and practice.^{9, 10} A more inclusive, connected agenda is urgently needed to generate the evidence to truly understand, monitor and respond to the interconnections between climate change and mental health.

Connecting Climate Minds

Connecting Climate Minds (CCM) is a Wellcome-funded project launched in 2023 to develop an inclusive agenda for research and action in climate change and mental health. The project has two key, intertwined aims. The first is to develop an aligned and inclusive agenda for research and action that is grounded in the needs of those with lived experience of mental health challenges in the context of climate change, to guide the field over the coming years. The second is to kickstart the development of connected communities of practice for climate change and mental health in seven global regions (designated by the Sustainable Development Goals), equipped to enact this agenda. We aim to combine the strengths of a global perspective and regional focus and bring together diverse disciplinary perspectives into a shared vision that can ensure research is effective at addressing priority evidence gaps and informing changes in policy and practice at the intersection of climate change and mental health.

Through bringing together experts across diverse disciplines, sectors and countries, the CCM team has facilitated the development of a lived experience-informed research and action agenda for the climate change and mental health field in Eastern and South-Eastern Asia (ESEA).

Objectives of the research and action agenda

The research and action agenda is designed to focus future efforts to help those who are experiencing, or will experience, the compounding mental health challenges of climate change. It aims to support those who are already responding to these challenges – through communities, research, policy and practice – by building a more connected and collaborative climate change and mental health field. It also aims to empower experts across disciplines and sectors to join and make progress in this area by identifying clear priorities and fostering a more inclusive and transdisciplinary field.

The agenda addresses these aims through three core objectives, which are to:

1. Identify priorities for research that can inform action to meet the needs of people experiencing and responding to the mental health impacts of climate change in ESEA.
2. Identify what is needed to appropriately conduct research and translate evidence to action in policy and practice in ESEA.
3. Build understanding among researchers, funders and policy experts across disciplines and sectors of their role in furthering climate change and mental health research and equip them with these clear and actionable priorities.

The regional agenda will be integrated with six other regional agendas to inform a **global research and action agenda** for climate change and mental health. This will ensure that global research efforts and investment in climate change and mental health are grounded in regional-level priorities. Importantly, the global agenda will also integrate insights from agendas developed with and for some of the most affected groups globally, namely Indigenous Peoples, youth, small farmers and fisher peoples.

Use of the terms climate change and mental health

Climate change, mental health, and their intersection are complex and wide-ranging fields. For the purpose of this agenda, we define the scope of these terms as follows.

By **mental health challenges**, we mean thoughts, feelings and behaviours that affect a person's ability to function in one or more areas of life and often involve significant levels of psychological distress. This includes, but is not limited to, anxiety, depression, post-traumatic stress, psychosis, suicidal thoughts, and substance misuse.

By **experiences of the effects of climate change**, we mean: 1) experiencing direct impact of climate hazards, such as more frequent and intense heatwaves, wildfires/bushfires, drought, floods or storms (e.g., typhoons, hurricanes, cyclones), and 2) experiencing disruption to the social and environmental determinants of good mental health, such as being forced to move home, not being able to access food or water, losing livelihood or homelands, or disruption to cultural practices because of climate change.

Mental health challenges in the context of climate change include:

- How climate change may lead to worsening pre-existing mental health challenges.
- How climate change may contribute to prevalence or impact of existing mental health challenges.
- How climate change may impact treatment access or effectiveness for those with mental health challenges.
- How climate change may lead to new mental health challenges.

The convening work of CCM presents a key opportunity to build our understanding of diverse perspectives, framings and terminologies in ESEA, which we have sought to reflect within the research and action agenda.

Background to Connecting Climate Minds

Regional Community of Practice

In ESEA, CCM is led by a Regional Community Team (RCT), responsible for convening diverse expertise across the region and building regional capacity to create and enact the research and action agenda. The structure of the RCT is outlined below.

Regional Community Convenor (RCC)

Purpose: Responsible for developing and delivering project activities in the region, including convening and supporting a regional community of diverse expertise.

Members: Renzo R. Guinto, John Jamir Benzon Aruta, Gail S. Co, Kent Tristan Esteban, Dianne J. Cunanan, Alfredo Lorenzo R. Sablay, Jun Angelo Sunglao [Planetary and Global Health Program, St. Luke's Medical Center College of Medicine - William H. Quasha Memorial, Philippines](#)

Co-convenors

Purpose: Bringing additional breadth of expertise across disciplines and countries (i.e. organisations spanning climate expertise, stress neuroscience and mental health expertise, in different sectors), providing technical advice and review, and supporting project delivery.

Members: Rodne Galicha, John Leo Algo, [Living Laudato Si' Philippines](#)

Joji Sugawara, Niaya Harper Igarashi, [Health and Global Policy Institute, Japan](#)

Nur Febriani, Maria Puspa Kartika, [Alam Sehat Lestari, Indonesia](#)

Laurence L. Delina, [Division of Environment and Sustainability, The Hong Kong University of Science and Technology](#)

Ramon San Pascual, Laetania Belai Djandam, [Health Care Without Harm- Southeast Asia](#)

Lien To, [Center for Health Environment Research and Development, Vietnam](#)

Lived experience advisory group (LEAG)

Purpose: Advisory board of experts with lived experience of mental health challenges in the context of climate change and/or belonging to vulnerable population groups and living with climate hazards. Drawing on their unique expertise and wisdom, LEAGs provide vital community-centred perspectives and guidance that inform the overarching approach and outputs of the project.

Members: Veronica “Derek” Cabe, Coal and Nuclear-Free Bataan Movement, Philippines

Marinel Ubaldo, Philippines

Laily Lutfiana Dhia, Alam Sehat Lestari, Indonesia

Madihah Ahmad Puaad, Angsana Health, Malaysia

Muhammad Daniel Azlan Mahadzir, Adeante Corporation, Malaysia

Jhonatan Yuditya Pratama, Patamuan Talino Foundation, Indonesia

Youth ambassador (YAs)

Purpose: Youth advisors (aged 18-29) with lived experience of mental health challenges in the context of climate change and/or belonging to vulnerable population groups and living with climate hazards. YAs bring unique youth-centred perspectives to the development and implementation of project activities.

Members: Jasmin Irisha Jim Ilham, Malaysia

Led by the Planetary and Global Health Program of the St. Luke's Medical Center College of Medicine in the Philippines serving as the RCC for ESEA, and together with the co-convening organisations, the LEAG, and the YA, the regional community aims to bring together a rich network of expertise and experiences from diverse backgrounds, united in a common interest in advancing the field of climate change and mental health in the region.

Methodology

We produced this research and action agenda through a robust, inclusive, and boundary-breaking methodology to capture, combine, and refine a rich diversity of perspectives while fostering connection across a growing community of practice.

The CCM core team developed this methodology in consultation with Regional Communities of Practice (RCOPs), a Global Advisory Board and Wellcome. Methods and materials were adapted regionally to ensure a balance of global standardisation with regional appropriateness and flexibility. Continuous sharing between regions of processes, learnings, and challenges facilitated iterative development of the methodology. The process for developing the regional research and action agendas is shown below.

Figure 1: Research and action agenda development methodology



Pre-dialogue scoping

Global scoping and framing

- We performed a global-level scoping review of current reviews, key papers and policy reports relating to the climate change and mental health field.
- We undertook a mapping of the research categories covered by and used to structure previous reviews of the field.
- We conducted a second mapping of the recommendations for action on climate change and mental health proposed by reports written to inform policy and practice.
- We used these results to frame the dialogues and research agenda to align with the current field while responding to key gaps.

Scoping research

- At the regional level, we conducted a pre-dialogue scoping which involved a rapid review of existing scholarship, international reports, relevant websites and policy documents. Five main areas were examined: regionally relevant climate hazards; climate-related mental health impacts documented or studied; national acknowledgement of the climate change-mental health nexus in national policies; initiatives around education, capacity-building and funding; and research activities on the nexus.
- We also conducted semi-structured interviews with experts from the region, including those from academia and those who work with communities and youth, especially those who could not participate in any of the dialogues. These interviews helped the team to gain a sense of the current state of play of the climate change-mental health nexus, identify any gaps in the current knowledge base, and explore potential areas for research and action.
- We engaged the regional co-conveners, members of the LEAG, and the YA, through several online meetings and offline through emails and shared documents. These collaborators helped define the scope of the investigation, provide information and resources that are not available online, and

identify stakeholders in the region that can be consulted and invited to participate in the dialogues.

- We also engaged with persons with lived experience of climate change and/or mental health (particularly members of the LEAG) through recorded storytelling sessions. This is to ensure that the production of the research agenda was appropriately grounded in an understanding of the key lived experience needs emerging in the region. The recorded lived experience stories are also used for CCM's advocacy and communication activities.

Dialogue methodology

- We held two virtual dialogues (3 hours each) with experts across disciplines, sectors and countries in ESEA (see Appendix 1 for agendas).
- Dialogue discussions were designed by the global team, amended to be locally appropriate, and facilitated by members of the RCoP, who are themselves research, policy, and lived experience experts.
- Many facilitators actively contributed to the discussions and therefore played a role in both data collection and production.
- The first dialogue identified regional needs and generated research priorities.
- The second dialogue gathered feedback on draft research priorities, identified how to enact research in the region and translate evidence to action in policy and practice, and explored the diversity of regional perspectives and understandings of key relevant concepts.
- Data generated in the dialogues included: Google Jamboard notes written by participants, Zoom chats, notes made by dedicated notetakers and transcripts of all discussions.

Survey methodology

- We distributed a **pre-dialogue online survey** prior to Dialogue 1 to inform dialogue design and to solicit perceptions on regional climate impacts, climate-related mental health impacts, and research priorities.
- We distributed a **post-dialogue online survey** after Dialogue 2 to obtain a second round of feedback on research priorities and to identify relevant methods, metrics, and datasets to address these priorities.

Analysis methodology

- **Research categories:** The Climate Cares Centre conducted a global landscaping exercise of relevant existing climate change and mental health reviews and identified four broad research categories as areas of critical need for further work globally. This framework was used as the basis for structuring discussions within dialogues to generate research priorities and formed the global coding framework for analysis.
- **Generating priority research themes:** Participants in Dialogue 1 (see Appendix 1 for agenda) were led through a structured discussion to surface their views on the emerging and likely mental health consequences of current and future regionally-relevant climate hazards, and opportunities for mental health benefits of action. Participants then generated draft research themes based on identifying where evidence would usefully inform responses in policy and practice.
- **Analysis:** Dialogue data (transcripts of breakout rooms and notes) was analysed using the Framework Method¹¹ – a matrix-based approach that allows qualitative researchers to undertake

deep interrogation of transcripts and written notes. The analyst team populated a matrix based on the global coding framework by creating nuanced summaries with key quotes drawing from the notes and transcripts of the break-out discussions.

- **Draft priority research themes:** This matrix was used to draft a list of priority research themes, refined through consultation with the RCoP and triangulation across breakout notes, transcripts, pre-dialogue scoping and expert consultation. Selection of priority research themes adhered to a globally developed structure and selection criteria (e.g., potential to answer greatest regional emerging needs and evidence gaps, potential to inform decision making in policy and practice, research feasibility; for full detail refer to Appendix 2).
- **Refinement of priority research themes:** Research themes were shared with dialogue 2 participants who gave feedback in response to the following prompts: 1) what would be the most helpful to know about this research topic to understand and respond to it?; and 2) what would you want to know first? Research themes were refined in response to this feedback and shared with dialogue participants and a wider sample of experts in the post-dialogue survey. Participants gave free text responses to the question '*Is there anything you would like to change, remove or add to this research theme?*' and were given the option to suggest additional research themes.
- **Finalisation of priority research themes:** A final list of priority research themes was generated based on incorporation of survey feedback, consultation with the RCoP and regional experts, CCM core team, Global Advisory Board and Wellcome.

Coding frameworks for the research agenda and action agenda can be found in Appendix 3 and Appendix 4, respectively.

Participants

Dialogue participants were a diverse group across geographical spread, gender, sector, and discipline. All participants were invited to both dialogues; however in some cases participants were unable to attend both dialogues.

In total 39 participants attended Dialogue 1 and 32 participants attended Dialogue 2. The tables below provide a breakdown of participant characteristics.

Geographical spread

Country	Dialogue 1		Dialogue 2	
	Number	Percentage	Number	Percentage
Australia	1	3%	1	5%
Brunei Darussalam	1	3%	1	5%
Cambodia	1	3%	0	0%
Hong Kong	1	3%	0	0%
Indonesia	6	16%	3	14%

Japan	1	3%	1	3%
Malaysia	5	13%	3	14%
Mongolia	1	3%	0	0%
Myanmar	1	3%	0	0%
New Zealand	1	3%	1	5%
Philippines	9	24%	6	27%
Singapore	2	5%	2	9%
Sweden	1	3%	1	5%
Taiwan	1	3%	1	5%
Thailand	1	3%	0	0%
Timor Leste	1	3%	1	5%
United Kingdom	1	3%	1	5%
Viet Nam	2	5%	0	0%

Expertise

	Dialogue 1		Dialogue 2	
Expertise	Number	Percentage	Number	Percentage
Climate Change	23	34%	12	30%
Mental Health	13	19%	9	22%
Healthcare	19	28%	11	28%
Other	11	16%	8	20%
I do not know / Prefer not to say	1	1%	0	0%

Discipline

	Dialogue 1		Dialogue 2	
Discipline	Number	Percentage	Number	Percentage

Activism	15	11%	9	12%
Community	23	16%	11	15%
Education	17	12%	6	8%
Expert through my own lived experience	13	9%	6	8%
Funding	4	3%	6	8%
Healthcare	10	7%	6	8%
Non-governmental Organisation	23	16%	11	15%
Policy	13	9%	5	7%
Research	22	15%	13	18%
Other	3	2%	1	1%

Gender

	Dialogue 1		Dialogue 2	
Expertise	Number	Percentage	Number	Percentage
Men	12	32%	8	36%
Women	24	63%	13	59%
Non-Binary	2	5%	1	5%

Number of survey participants:ⁱ

Pre-dialogue survey: 71

Post-dialogue survey: 28

Ethics, data collection and storage

This study has been reviewed and given an ethical favourable opinion by the Imperial College Research Ethics Committee (Study title: 'Global Dialogues to set an actionable research agenda and build a

ⁱ Please note numbers are approximate and do not account for duplicate or incomplete responses.

community of practice in climate change and mental health', study ID number: 6522690). Details on data collection and storage can be found in Appendix 5.

Current state and emerging needs for climate change and mental health in Eastern and South-Eastern Asia

As the climate crisis escalates, more people globally are experiencing related mental health consequences. However, the current evidence base doesn't fully capture these experiences. To develop research themes that ultimately meet the needs of those experiencing and responding to the interconnections between climate change and mental health, it is vital to know: 1) what do people from different backgrounds, contexts, and sectors – particularly those with lived experiences of mental health challenges in the context of the climate crisis – report as their experiences, needs, and resiliencies, and 2) what evidence do people making decisions and taking actions on the ground need in order to adequately respond.

This section sets out the context of the research agenda, presenting a synthesis of what we heard through the dialogues, surveys, expert consultations, and literature review as key emerging needs for mental health in the context of climate change. We also present regional perceptions and insights from the literature on the current state of evidence to explore which key emerging needs are not yet being captured through research.

Current research on climate change and mental health in Eastern and South-Eastern Asia

Overall, there is a scarcity of research on the climate-mental health nexus in Eastern and South-Eastern Asia, especially research conducted by scholars and practitioners from the region. However, the slowly emerging scholarship on the mental health consequences of climate change in the region reveals that people who are most affected by climate hazards (e.g., young people, the elderly, disaster survivors, farmers, etc.) experience trauma, depression, anxiety (including the new phenomenon called 'climate anxiety') and sleep problems, among others. Countries where there is emerging scholarship on this topic include China, Japan, Philippines and South Korea. ^{12, 13, 14, 15}

Meanwhile, based on our pre-dialogue scoping (i.e., rapid review of literature, policies, interventions and programs related to climate change and mental health), national policies and programs related to either climate change or mental health in the countries in the region do not explicitly acknowledge the intersection between climate change and mental health. While efforts to tackle climate change are increasing in the region, the inclusion of mental health aspects in climate policies and programs remains limited. For instance, [Hong Kong's Climate Action Plan 2050](#) details plans and strategies to increase disaster resilience but psychological and mental resilience amid climate change was not mentioned. Similarly, the conceptualisation of mental health policies and programs in countries throughout the region has yet to recognise the direct and indirect role of climate change in people's mental health. For example, the Mental Health Act in the Philippines which aims to establish a national mental health policy for enhanced mental health services only mentions the provision of psychosocial support in the context

of disasters, but not through the lens of climate change.¹⁶ Nonetheless, existing regional and domestic efforts on disaster risk reduction and management, especially when it comes to emergency response and rehabilitation, already incorporate the provision of psychosocial support to disaster victims;¹⁷ expanding this to include a broader climate lens is an important step forward in advancing this new agenda. Moreover, there are a few promising initiatives (e.g., awareness raising, community-building, education, training and pilot research) in various countries in the region that are starting to give a spotlight to the climate-mental health intersection, such as the inclusion of this topic in the [Southeast Asia Climate and Health Responder Course](#) offered in 2023 and national conference of the [Psychological Association of the Philippines](#) in 2022. More support must be given to these nascent activities in order to enhance the region's efforts in addressing climate change and its inimical impact on people's mental health.

Framing of key concepts

The overall scope and focus of CCM has been guided by the framing of climate change and mental health outlined in the introduction. 'Climate change', 'mental health', and their intersections and related terms, along with other relevant key concepts, are also understood and defined in diverse ways in ESEA. Identifying, acknowledging and honouring the ways these terms are understood and used in different settings is critical to help foster connections, awareness, and recognition across disciplines, cultures, and communities. This section highlights regionally relevant understandings, as generated through a mapping activity in Dialogue 2, designed to reveal the diversity of perspectives around key concepts and other discussions throughout the project.

Mental health

To understand the spectrum of definitions and concepts that the participants view as related to mental health, it is important to note that many countries in ESEA historically endorse traditional and conservative views on mental health that influence the public's overall perspective on mental health.¹⁸ ¹⁹ For instance, symptoms of mental health problems are historically associated with possession of evil spirits and bad luck, which lead to stigma and harsh treatment of individuals showing mental health symptoms. In recent years, mental health information campaigns in the region, especially online and via social media, have helped to raise awareness and reframe views about mental health.²⁰ Dialogue participants are educated individuals and professionals, which may explain their generally informed views about mental health.

The first word for the spectrum mapping activity was 'mental health' (Figure 2). Participants associated mental health with a plethora of concepts, including wellbeing, peace, sanity, fulfilment, contentment, stability, community and spirituality. Common definitions of mental health included having a "state of mind" that enables people to cope with life's stressors effectively and live life to its fullest. It was also related to living in a safe and comfortable environment, which supports an individual's day-to-day function (e.g., clean house, food, etc.), as well as forging healthy and meaningful relationships with people around them. The [Philippine Mental Health Association](#) and [Malaysian Mental Health Association](#) are example organisations in the region that promote these broad and holistic views of mental health beyond the absence of mental illness.

Figure 2: Regional spectrum mapping of the phrase ‘mental health’



Resilience

On the word ‘resilience’ (Figure 3), participant responses were more aligned, especially in terms of the word’s definition. A common view on the definition of resilience was the ability to adapt and cope with stressors, while having the capacity to not only recover, but also to improve and build back better as a result of such stressors. “Adapt, endure, thrive” and “bending without breaking” were some of the statements offered during the dialogue. Most statements given by participants are phrased in such a way that they can be applicable to either climate change or mental health: two fields that use the term ‘resilience.’ Nonetheless, in several countries in the region, academic institutions frame the term ‘resilience’ in the context of disasters including those caused by climate change, such as the case of the [University of the Philippines’ Resilience Institute](#) in the Philippines and [The National Research Institute for Earth Science and Disaster Resilience](#) in Japan. Participants also offered more ‘societal’ views of resilience, such as this concept being associated with the complacency of leaders in addressing people’s needs. This stems from the notion that certain populations are believed to be able to adapt and survive challenges – in short, resilient – despite the lack of proper interventions addressing their root problems. This is why it is important that resilience is not used as “an excuse for policy sectors to not provide basic needs” and that “electing the right people brings a more resilient population.”

Figure 3: Regional spectrum mapping of the word 'resilience'



Climate Justice

'Climate justice' (Figure 4) was associated with ideas around climate activism and holding major climate change contributors accountable. "Making rich polluters pay" and "be the voice for the voiceless" are examples of statements that emerged during the dialogue. Calls for solidarity were also raised during the mapping exercise, such as the amplification of youth voices and protection of vulnerable and marginalised groups (e.g., environmental activists), with an emphasis on human rights. Transition into cleaner and greener energy that is accessible to all, decolonisation of climate justice, just compensation for climate victims and a stance against greenwashing – deceptive advertising practices making companies or products seem more environmentally friendly than they are – were also highlighted during the dialogue. These ideas are consistent with the growth of the climate justice movement within the region. For example, organisations such as [Climate Justice Indonesia](#), which calls water pollution, air pollution and deforestation "Indonesia's killers," and [Philippine Movement for Climate Justice](#) demand accountability from the world's major polluters.

Figure 4: Regional spectrum mapping of the phrase 'climate justice'



Regional needs for mental health in a changing climate

While the climate crisis affects every part of the world, current and future hazards are unevenly distributed and variable. For instance, coastal communities in the Pacific face sea level rise, coastal erosion, and storm surges, while Inuit communities face melting sea ice. Different parts of the same country or region may face different levels of threat to droughts, wildfires, floods, or extreme heat. Understanding the predicted climate hazards and their timescales for different communities and regions is vital to appropriately target support for mental health.

This section outlines the key climate exposures facing ESEA projected to approximately 2030, as modelled by climate experts at the Red Cross Red Crescent Climate Centre. We also present findings from the dialogues, surveys, and consultations with key stakeholders in the region to explore perceptions of: 1) mental health risks associated with these climate exposures, 2) whose mental health may be most at risk, 3) the pathways through which climate exposures might produce or exacerbate existing mental health challenges, 4) climate adaptation and mitigation actions that may benefit mental health, and 5) mental health actions/solutions that can help respond to these mental health impacts.

Climate hazards

ESEA is facing an increase in the frequency and severity of a range of climate hazards, including the following, modelled to approximately 2030 as compared to historical baseline (generally 1986-2005):^{21, 22, ii}

ⁱⁱ Future projections are based on the middle of the road emissions scenario (SSP2-4.5 Shared Socio-economic Pathway) from the CMIP6 multi-model ensemble provided in the IPCC, 2021.

- **Extreme temperatures**, with up to 40 additional days above 35 C per year in Thailand, Indonesia and parts of China (high confidence);ⁱⁱⁱ
- **Droughts** across the region (low confidence);
- **Wildfires** in northern expanses of the region (low-to-medium confidence);
- **Heavy precipitation**, including an increase in extreme precipitation events and average precipitation across the region (high confidence);
- **Flooding** in Northern Asian regions and prolonged inundation in the Mekong Delta (high confidence);
- **Sea level rise** associated with coastal area loss, shoreline retreat and coastal flooding (high confidence); and
- **Tropical cyclones**, with an increase in frequency (approximately 6%), yet a decrease in intensity of Category 4-5 tropical cyclones in this region (low confidence).

Every year, these climate-related events increasingly cause losses of thousands of lives, devastations of livelihoods, displacement and loss of cultures, among others. While the extent of exposure to these threats varies across countries depending on their geographical location, economic capacity, and disaster preparedness, the consequences and losses to people's lives and livelihoods are increasingly and disproportionately experienced by vulnerable communities, especially in developing countries in the region.

What mental health outcomes appear to be impacted?

The mental health impacts of climate change repeatedly identified during the dialogues, surveys, and consultations with lived experience experts include depression, anxiety, post-traumatic stress disorder (PTSD) and suicidal ideation. Other mental health outcomes that were sparsely mentioned include substance abuse, bipolar disorder and schizophrenia.

Who is particularly affected?

Dialogue participants identified several groups particularly vulnerable to the mental health impacts of climate change, as listed below.

- Disadvantaged demographic groups
 - Children and youth
 - People living with disabilities
 - People with pre-existing medical conditions
 - The elderly
 - Women and non-binary gender identities
- Hazard-exposed occupational groups
 - Farmers
 - Fisher peoples
 - Outdoor workers

ⁱⁱⁱ Confidence levels listed after projections are a measure of researcher confidence that model's results are true and not due to chance.

- Working class individuals
- Geographically and socioeconomically disadvantaged groups
 - Coastal communities
 - Geographically isolated communities
 - Indigenous populations
 - People living in disaster-prone areas
 - Urban poor
- People in institutionalised settings
 - Prisoners
 - Those in immigration detention facilities
 - Those in residential mental health facilities and psychiatric wards
- Groups acting on climate change and its health impacts
 - Climate activists
 - Climate scientists
 - First responders
 - Health workers

What are the pathways and mechanisms linking these climate hazards to mental health outcomes?

An important discourse during the dialogues centred around the potential pathways and mechanisms that explain the link between climate change and mental health in ESEA, which has been largely unexplored in the literature and policy. Table 5 below provides an overview of the potential pathways and mechanisms that surfaced during the dialogues.

Table 5: Pathways and mechanisms linking climate hazards to mental health outcomes

Climate hazard	Population affected	Mental health outcomes	Mechanism
Climate-change related disasters (e.g., typhoons, floods, intense rains and winds, etc.)	People living in disaster-prone areas	PTSD and anxiety	Repeated experience of climate-change related events increase susceptibility to PTSD and anxiety.
	Women and non-binary gender identities	Depression and anxiety	Gender inequities exacerbated during disasters lead to increased violence and discrimination, which may result in depression and anxiety.
	Children	General mental health outcomes and cognitive development	Schools are converted into evacuation centers during disasters, disrupting education

		effects	among school children.
	Disaster survivors	Depression and anxiety	The inability to recover from the economic loss, destruction of properties, disruption of livelihoods leads to depression and anxiety, coupled with the limited access to basic needs following disasters, contribute to the depression and anxiety of disaster survivors.
	Socioeconomically disadvantaged groups	Depression and anxiety	Increased cost of living due to climate impacts and the inability to access coping technologies to mitigate climate-related stressors (e.g., air conditioners and air purifiers) lead to depression and anxiety.
Slow-onset events (e.g., environmental destruction, rising temperatures, rising sea levels, etc.)	Farmers and fisher peoples	Depression, anxiety, and suicidal ideation	Economic losses due to a decrease in the quality and quantity of crops, livestock, and fish catch may lead to depression and anxiety among farmers and fisher peoples whose livelihoods depend on the environment. Worst cases may also lead to suicide and suicidal ideation.
	Indigenous populations	Depression and anxiety	Indigenous Peoples who have a strong connection to nature and their land may experience depression and anxiety as the environment is affected by climate change.

What climate adaptation and mitigation actions have benefits for mental health?

Dialogue participants identified several climate adaptation and mitigation actions that potentially benefit mental health, ranging from individual and community initiatives to government responses and systemic improvements. These include:

- **Nature-based solutions**, including increasing tree cover and green spaces in urban environments, which not only offsets the effects of climate change (e.g., reduces the impact of rising temperatures) but also potentially improves mental health, as shown in some studies.²³
- **Sustainable urban planning**, such as the integration of green architecture, and urban adaptation measures, such as better sewage systems to reduce flooding and sustainable cooling materials for buildings to combat the extreme heat.
- **Disaster preparedness and response strategies**, including the planned and careful relocation of vulnerable populations in disaster-prone areas. Dialogue participants stressed the importance of considering the location of temporary settlements, along with socioeconomic characteristics (e.g., availability of livelihood opportunities).
- **Individual and community-level sustainable practices**, such as waste segregation at home and participation in community clean-ups, which can reduce anxiety by engaging in environmental actions. A vital point repeatedly raised was the importance of involving the whole community in climate action. Actions that are locally-contextualised, culturally-sensitive and people-inclusive can help build a sense of community that may improve the sustainability of climate actions and the community's overall mental health. The integration of local and traditional knowledge, especially from Indigenous communities, into climate action was also emphasised during the dialogue.
- **Government actions and systemic changes**, such as making healthcare systems climate-resilient and sustainable, which helps restore the environment while improving mental health promotion and service delivery to the population. Additionally, dialogue participants emphasised the importance of collaboration across governments, sectors, and agencies, which may not only accelerate climate mitigation and adaptation but also create the societal conditions that benefit mental health. The integration of mental health into climate change policies and vice versa and the careful and effective implementation of such integrative policies may also help create a social and political system that fosters care for both the health of people and the planet.

What mental health actions can help respond to these mental health impacts?

In order to design and deliver appropriate interventions, the unique cultural aspects of target populations, as well as their specific social, economic, and geographical conditions, must be taken into account.²⁴ This can only be achieved through close, intentional and meaningful collaboration across various sectors and with local communities. Participants highlighted the importance of incorporating traditional knowledge and community efforts into mental health interventions to ensure long-term sustainability. Disasters and other climate-related events were important points of discussion with an emphasis on the presence and improvement of mental health and psychosocial support (MHPSS) services during such events, both for disaster survivors and disaster responders. Increased exposure to nature and 'ecotherapy' were also acknowledged as both a mental health intervention and a potential motivator for participating in climate action.

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

Further contextual information on regional needs

Young people have always been identified as vulnerable to climate change, because of their limited adaptive capacity in their younger years but also much increased risk in their later years.²⁵ The youth population (ages 15 to 34) in ESEA continues to expand. For instance, a total of 750 million young people aged 15 to 24 years old are living in the East Asia and Pacific region, comprising about 60% of the world's youth population.²⁶ Specifically for Southeast Asia, the youth represents 34% (226 million) of the region's total population (666 million).²⁷ As today's youth will unfortunately be living in a future that will see the worst consequences of the climate crisis, the young people of ESEA must be given greater attention and included in today's climate discourse and decision-making.

A few upper middle- and high-income countries in the region (e.g., China, Japan, Singapore, South Korea, Taiwan) are currently major contributors to climate change. Many countries, on the other hand, are low- and lower middle-income countries (e.g., Cambodia, Indonesia, Laos, Philippines, Thailand, Vietnam, etc) with minimal responsibility in causing the climate crisis, especially in terms of the region's (and the world's) greenhouse gas emissions. However, these countries face the severe consequences of the climate crisis, which are amplified because of economic limitations. Acknowledging these differences is vital for achieving a holistic and nuanced understanding of the regional situation when it comes to the climate change-mental health nexus. It also may help build solidarity across countries and inform the development of fair and just solutions to both climate change and its impacts on mental health.

The region is rich in Indigenous and other traditional cultures, as the countries in the region are home to hundreds of Indigenous populations as well as numerous faith traditions - including Islam, Christianity, Buddhism, Confucianism, Zoroastrianism and others, including their different iterations. Currently, approximately 260 million Indigenous individuals representing 2000 distinct civilisations and languages reside in Asia.²⁸ Indigenous groups in the regions historically and currently face structural and systematic discrimination and injustices, as experienced through forced displacement, loss of ancestral lands and violation of human and cultural rights.²⁹ These injustices ultimately result in the decline of their population, rapid loss of their cultural identities, and possibly deterioration of their mental wellbeing. Climate action and mental health interventions must highly consider Indigenous perspectives, ensure their inclusion and respect their dignity and rights.

Research agenda

Priority research themes

Background to research categories and priority research themes

This research agenda presents an aligned vision to guide the climate and mental health field in ESEA. Research priorities have been generated through consultation with experts across disciplines, sectors and geographies in the region and iterated with experts regionally and globally. The priority research themes represent areas where targeted research investment could create a full picture of impacts, their mechanisms and solutions across both mental health and climate actions. We outline why these have been identified as priorities and how they can be addressed by combining expertise across disciplines and sectors.

Research priorities are presented within four overarching research categories that were identified as areas of critical need for further work globally and that map the climate and mental health research space at a high level, based on an initial review of global literature. Note that some priorities span multiple categories.

- **Impacts, risks and vulnerable groups:** improving our understanding of the extent to which mental health is affected by climate change and for whom. For example: what mental health outcomes are impacted or at risk; the prevalence, severity, economic and societal costs of these impacts; and who is most vulnerable to these impacts.
- **Pathways and mechanisms:** improving our understanding of how climate change affects mental health and, in particular, whether there are factors specific to climate change that increase mental health risks or create new experiences of mental health challenges. This includes considering biological, psychological, societal, or environmental pathways and mechanisms.
- **Mental health benefits of climate action (adaptation and mitigation):** understanding and quantifying when and how climate adaptation and mitigation actions, across sectors, can also have win-win benefits for mental health.
- **Mental health interventions/solutions/actions in the context of climate change:** identifying the most effective mental health interventions/solutions/actions to support mental health in the context of climate change, across diverse sectors. This encompasses providing support to people already experiencing negative mental health impacts and reducing risk or severity of future negative impacts.

We hope that these priorities will act as an inspiration to guide the direction of the research community of practice, provide investment targets for funders, and focus generation of evidence to best enable policy makers and practitioners to address the emerging and predicted mental health needs in response to climate change in ESEA.

Priority research themes

Research category	Priority research theme (n = number of research questions)
1. Impacts, risks and vulnerable groups	1. Understanding what mental health challenges (e.g., anxiety, depression, PTSD, suicide, etc.) people in the region experience when affected by extreme weather events (such as typhoons, flooding, bushfires, landslides and drought) and slow-onset climate events (such as sea level rise and increasing ambient temperature), how many people are affected and how affected populations cope and adapt to these challenges. 2. Through a health equity lens, investigating how specific groups are affected by worsening mental health symptoms and conditions as a result of both extreme weather and slow-onset events brought about by climate change. 3. Through a health equity lens, unravelling geographic (e.g., inter-country) and cross-population (e.g., between Indigenous and other populations) differences in the magnitude and manifestation of climate-related mental health symptoms and conditions and the factors driving these differences. 4. Identifying and evaluating existing assessment tools and developing new measures and tests (unique to the mental health responses to climate change) to screen, assess and diagnose the various mental health symptoms and conditions resulting from climate change-related extreme weather and slow-onset events, both in the general population and in highly vulnerable populations. 5. Understanding what mental health challenges (e.g., anxiety, depression, PTSD, suicide, etc.) people in the region experience when affected by extreme weather events (such as typhoons, flooding, bushfires, landslides and drought) and slow-onset climate events (such as sea level rise and increasing ambient temperature), how many people are affected and how affected populations cope and adapt to these challenges.
2. Pathways and mechanism	SOCIETAL 6. Understanding how socioeconomic inequities, worsened by climate change, and their accompanying access problems (in relation to basic needs, healthcare, climate adaptation technologies, etc.) affect the magnitude and intensity of mental health challenges. 7. Identifying the demographic and socioeconomic factors that enhance the psychological resilience of individuals and communities against the mental health impacts of climate change. 8. Understanding how climate change-related extreme weather and slow-onset events interact with gender-based inequalities to affect mental health outcomes, and what other social, economic and cultural factors influence these relationships. 9. Understanding how stigma impacts mental health outcomes and help-seeking behaviour among populations affected by climate change, whether this limits their access to mental health services during and after climate change-related disaster, and how such stigma can be addressed. 10. Understanding how the disruption of education during climate change-related disasters and slow-onset events results in or contributes to mental health symptoms and conditions such as anxiety and depression among children and young people.

11. Exploring how climate education/awareness/literacy impacts mental health, especially in young people, and how such education can improve mental health and drive climate action.

ENVIRONMENTAL

12. Understanding the proximal and intermediary factors in the climate hazard-human health pathway that increase or decrease people's vulnerability to mental health symptoms and conditions such as irritability, loss of hope, loss of concentration, anxiety and depression.
13. Understanding how urbanisation (including associated air pollution) and ecosystem destruction interact with climate change to produce psychological reactions such as ecological grief, solastalgia and climate anxiety, and whether and how these reactions lead to or exacerbate mental health challenges.

PSYCHOLOGICAL

14. Understanding how the experience of climate change interacts with exposure to social media, including climate change-related content, in influencing the mental health of children and young people.
15. Understanding how the experience of climate change interacts with generational trauma experienced by children and young people in influencing their mental health.
16. Identifying which pre-existing mental health conditions are more likely to be affected by climate change-related events and how.

BIOLOGICAL

17. Understanding the extent to which various climate hazards (and the alterations they create in proximal factors) influence neurological (including brain development), endocrine, aging, chronic disease, psychological or genetic (i.e., epigenetics) processes that may lead to adverse mental health outcomes.
18. Exploring how climate change-related impacts on diets and food systems (i.e., food quality, food scarcity, etc.) affect mental health outcomes.

3. Mental health benefits of climate action (adaptation and mitigation)

19. Measuring the mental health benefits of climate mitigation and adaptation efforts at both the individual and population levels, and how these can be incorporated into policy decisions to enhance climate policies.
20. Measuring the mental health benefits of exposure to greener and bluer environments resulting from climate mitigation/adaptation efforts and whether and how direct experiences of these benefits can drive individuals to take climate action.
21. Understanding how being part of climate-resilient communities and local climate mitigation and adaptation efforts impacts mental health symptoms and status.
22. Understanding whether and how climate-resilient and environmentally sustainable healthcare systems could improve mental healthcare access and reduce mental health challenges.
23. Analysing the development, integration and implementation of climate change and mental health policies and programs at local, national and regional levels.
24. Exploring how different stakeholders (including governments, civil society, academia, private sector and communities with lived experiences) and disciplines can meaningfully collaborate to advance joint learning, data sharing, research, policy and intervention at the climate change-mental health nexus at local and national levels.

4. Mental health interventions/solutions in the context of climate change

25. Exploring how to strengthen mental health and psychological support (MHPSS), especially at primary care level, to address mental health impacts of climate-related disasters and slow-onset events.
26. Exploring strengths-based approaches (e.g., positive psychology and positive education) to alleviate mental health challenges driven by climate hazards.
27. Identifying and evaluating community-based mental health interventions to address future mental health impacts of climate-related disasters and slow-onset events, particularly in climate-vulnerable and climate-affected areas. In particular, understanding how these interventions can be improved and tailored for climate change-affected communities.
28. Exploring how to incorporate cultural beliefs and practices from various faith communities and Indigenous populations in the region, including spirituality, into mental health interventions addressing climate change-driven mental health impacts, in accordance with a context-based and decolonised approach to mental health.
29. Exploring how collective actions and a greater sense of community affect climate change-induced mental health impacts and/or promote emotional coping and adaptation to climate awareness and experiences.

Overview of themes for impacts, risks and vulnerable groups

Why were these themes chosen as priorities?

These foundational research questions were glaringly missing in the limited scientific literature in the region and explicitly highlighted by dialogue participants and interviewed experts. Before effective interventions and responsive policies can be developed, we must establish a clear understanding of the phenomena being addressed. There are many unknowns with regards to which climate hazards lead to which types of mental health outcomes, and which subpopulations are vulnerable to such climate-induced mental health impacts. The limited attention given to the climate change-mental health nexus, especially in national climate and mental health policies, can be attributed to the lack of clarity of awareness and evidence on these interconnections. Dialogue participants emphasised the urgency and scale of the climate change and mental health crisis, hence the demand for more epidemiological studies measuring its magnitude. They also acknowledge climate change and mental health as issues of health equity, recognising that different populations are affected by these problems in different ways and degrees. The identification of vulnerable groups and the measurement of climate-induced mental health inequalities were also deemed utmost priority for research.

What are examples of methodologies, metrics and datasets that could be used to address these?

Collecting new data and accessing existing datasets are key to understanding the current state of mental health impacts of climate change in the region. Dialogue participants and individual expert interviewees shared the importance of linking all available local, national and regional climate and environmental data produced by environmental and meteorological bureaus with mental health and psychiatric data coming from hospitals, clinics, and national health agencies. However, it was also pointed out that these data

may not be readily available and accessible in certain countries and subnational jurisdictions, which poses a challenge to gathering and integrating climate change and mental health data. Responses from dialogue participants and findings from the pre-dialogue scoping also revealed the need to track data on climate-related hazards (e.g., heat, drought, storms, sea level) in parallel with mental health data over time (i.e., time-lagged) to understand potential fluctuations of mental health challenges alongside changes in climate events and hazards across time. Other data sources identified during the dialogue and pre-dialogue scoping include online news and social media content that reflect climate change narratives and people's emotional responses to climate-related events (i.e. sentiment analysis) which can provide insights on the linkages between climate change and mental health. Big data and modelling approaches, which are often used by climate modelers, can be used to predict future mental health in relation to climate trends, including the prevalence of mental health conditions and incidence of mental health-related events such as suicide.

To determine the prevalence of mental health issues associated with climate change, dialogue participants and individual expert interviews identified the use of quantitative methodologies such as observational epidemiological approaches, as well as quasi-experimental approaches to establish cause-and-effect relationship between climate variables and mental wellbeing. There is also a need to employ in-depth and qualitative approaches such as case studies, interviews, focus groups, ethnography and participatory approaches to understand the nuances of people's experiences of climate hazards and their mental health reaction to events caused by the climate crisis.

Overview of themes for pathways and mechanisms

Why were these themes chosen as priorities?

The themes represent the most common and frequently mentioned pathways and mechanisms potentially explaining the link between climate change and mental health in the ESEA region. These themes were selected as priorities because they can increase our understanding of the complex and unique ways climate change is affecting the mental health of people in ESEA, which is largely unknown based on the current status of scientific literature. Social pathways and mechanisms identified by Dialogue 1 participants as particularly important to the region include: climate change exacerbating existing socioeconomic inequities, especially in low-income and middle-income countries; gender-based inequality and violence; and stigma that causes barriers to care seeking in climate-vulnerable populations. There are also psychological, environmental and biological pathways and mechanisms highlighted by Dialogue 1 participants as mentioned in the previous section.

These research themes can serve as a guide in research prioritisation to ensure nuanced understanding of how climate hazards affect the mental health of populations and groups in the region. Mental health interventions in the context of climate change cannot be effective without sufficient understanding of the pathways in which climate change worsens mental health problems.

What are examples of methodologies, metrics and datasets that could be used to address these?

Understanding the different pathways and mechanisms would require utilising existing datasets including income and gender inequality data at the community and public health data (health and mental health) at national levels in ESEA. Results of Dialogue 1 emphasised that inequalities in the context of climate change are worse in societies where other forms of inequalities such as for different incomes and genders are present. Participants suggested that this could result in more severe mental health problems for these

groups. Additionally, climate modelling data (heat, storms, rainfall, etc.) can also be used to associate climate events with specific societal mechanisms such as violence and conflict and discrimination, among others, which lead to mental health problems, especially among vulnerable groups (e.g., people with pre-existing medical and mental conditions, persons with disabilities, low-income groups, children, etc.). Biological data, such as hormone levels and histological data, are also needed to understand how climate change is affecting the biological processes including neurological processes, which could result in mental health problems. Importantly, all these data must be examined longitudinally (after a certain number of months depending on the weather changes, annually or after several years interval, etc.). We note that understanding the various pathways and mechanisms would also require qualitative approaches to gain an in-depth appreciation of the lived experiences of people whose mental health problems and vulnerabilities are exacerbated by the climate crisis.

Overview of themes for mental health benefits of climate action [adaptation and mitigation]

Why were these themes chosen as priorities?

Expert participants emphasised that it is essential to acknowledge climate change as one of the causes of the mental health crisis, and we must prioritise examining how engaging in climate mitigation and adaptation could help reduce symptoms of mental health problems in the region. Prioritising these research themes on mental health benefits of climate mitigation and adaptation can help determine what type of efforts can improve the mental health and wellbeing of specific groups and populations (e.g., Indigenous Peoples, individuals with low-income backgrounds, children, persons with disabilities, etc.) in the region. Participants also highlighted that these themes reflect the serious and urgent environmental problems that are linked with the mental health issues in the region. Dialogue participants and experts also noted that, conversely, stable and improved mental health could be precursors for greater engagement in climate action and environmental protection.

What are examples of methodologies, metrics and datasets that could be used to address these?

Participants in Dialogue 1 and the convening team in ESEA acknowledge the importance of qualitative methods as powerful tools to assess how engagement in climate mitigation and adaptation can improve mental health outcomes, perhaps via mechanisms such as a perceived increase in safety and wellbeing. For example, case studies, participatory approaches and focus group discussions, among others, can help in improving our understanding on how climate mitigation and adaptation can help minimise the life and mental health threatening impacts of climate change events and create a safe environment for communities especially in times of climate disasters and catastrophes. Experts in the region also emphasised that past and current climate mitigation and adaptation initiatives must be evaluated on whether they also helped improve community stability, physical and mental health, and overall wellbeing.

Further understanding the link between climate change and mental health in the region would also require the use of big data. For instance, researchers can utilise datasets from clinics and hospitals to glean insights on the number of psychiatric intakes in communities and their exposures to climate hazards. Such datasets can also allow researchers to analyse whether communities who engage in climate action report improved mental health. Additionally, as evidence shows that climate variability is linked with increased crime rates,³⁰ it would be useful to analyse community- and country-level exposure to climate hazards, crime rate reports and mental health data. Comparing police records of crime rate and

mental health problems among severely climate-inflicted communities can also help generate important evidence and insights on the social and ecological mechanisms that link climate change and mental health outcomes. Moreover, it is also necessary to review and assess to what extent public institutions (e.g., hospitals, police, disaster response units and other government units) have already integrated climate change into their frameworks and operations. This can give valuable information on how much more effort is needed to make sure that communities and societies in ESEA are climate-aware, as it is important not only for addressing climate change but also improving people's mental health.

Overview of themes for mental health interventions/solutions in the context of climate change

Why were these themes chosen as priorities?

A recurring problem identified from the expert interviews and the dialogues is the disconnection between mental health and climate change, especially in the current efforts to address these issues. It is important to bridge this gap by exploring how climate change considerations can be incorporated in current mental health interventions, and conversely, considering mental health in current climate actions. For instance, delivering proper mental health and psychological support (MHPSS) is essential during climate change-related events to alleviate the mental stress that vulnerable groups undergo. In the context of climate-related disasters such as typhoons and floods, disaster survivors and disaster responders are especially vulnerable to mental health impacts in addition to physical health issues. Therefore, prioritising research on the feasibility and effectiveness of mental health response during disasters that takes into account aspects of climate change is critical. Research should explore whether mental health interventions are tailored to the needs and characteristics of the target community and truly benefit the involved vulnerable groups. Given the lack of awareness among the public on climate change and mental health, investigating educational and communication interventions that may help mainstream these topics and encourage more open discussions, especially in community settings, is another key research priority.

What are examples of methodologies, metrics and datasets that could be used to address these?

Exploratory studies and qualitative methods are potentially effective ways to explore themes related to mental health interventions. Conducting scoping studies with communities ensures that interventions and policies resulting from such investigations are appropriate for the groups involved. One way to achieve this is by capturing the lived experiences of community members who are affected by climate change-induced mental health outcomes, as well as those who are already receiving existing mental health interventions that need to be refined or tailored for climate-affected settings.

Mental health and psychological support during disasters should also follow a framework that is contextualised to the locality affected while considering a broader climate lens. Screening and assessment tools that are culturally sensitive and tailored to communities' varying characteristics should be developed to better capture their experiences.

There are existing frameworks from implementation research and intervention science that can be used in studying mental health policies and interventions.³¹ These frameworks can be further enriched by applying a climate change lens, for instance in terms of assessing current mental health actions based on their adaptability/responsiveness to the effects of climate change. Adopting an iterative process of

evaluating interventions and using insights to refine them allows for continuous improvement, enhanced responsiveness, and longer-term sustainability.

Action agenda

As the climate change and mental health field in ESEA continues to evolve and the evidence base grows, it is crucial to avoid perpetuating existing challenges, including: 1) disconnections across disciplines and between researchers and policy makers; 2) unequal focus on topics and geographies; 3) and siloed decision making for climate and for mental health. This action agenda sets out a shared vision as a rallying focus of the mental health and climate change field in ESEA. This includes guidance on how to best support the growing community of practice, how to translate evidence to action and the principles that should guide this approach. Enacting this agenda will require transdisciplinary effort and coordinated action across research, research funding, policy and practice. This action agenda aims to guide this work by setting out the challenges which must be addressed, opportunities that can be harnessed and priority actions to work towards a thriving climate and mental health field.

Regional vision for mental health in a changing climate

Vision statement

Overall, our vision is an ESEA region where people live and thrive – not only physically but also mentally and emotionally, in spite of the climate crisis – while also harnessing their personal and community resilience to drive action to tackle the climate crisis. Achieving this vision requires multiple components. The first is to establish and strengthen collective understanding of the interconnection between climate change and mental health, profoundly informed and enriched by accounts of people's lived experiences of climate hazards and their mental health consequences, and through consultations, dialogues, research and advocacy that brings stakeholders from across geographies, sectors, disciplines and generations. This regional vision centres the diverse knowledge and experiences of various communities and puts them at the heart of collaborative research and collective action. The vision is also committed to not only diagnosing problems but driving solutions that benefit people's lives, especially the vulnerable and the marginalised. Central to achieving this vision is the growth, expansion and sustainability of the Regional Community of Practice that emerged from this project, which will advance the climate and mental health field and actively advocate for safeguarding the mental health of vulnerable communities across the region.

We envision the future of climate change and mental health in Eastern and South-Eastern Asia as one that is guided by:

- Local wisdom, ensuring that Indigenous knowledge and traditional practices inform the design and delivery of mental health solutions especially in communities affected by climate change;
- Lived experience, especially of those who are the most vulnerable to climate-related events, ensuring that interventions are carefully designed and equitably implemented to serve those who need them the most;
- Youth, as their passion and innovation can drive new solutions to age-old problems at the intersection of climate change and mental health;

- Transdisciplinary and cross-sectoral collaboration, as the climate change-mental health nexus is a problem whose complexity requires the co-creation of innovative solutions that cuts across various disciplines and sectors of society; and
- Cultural and geographical inclusivity, ensuring that voices from all corners of the region are heard, are represented and have a stake at both local and regional decision making.

Creating an enabling environment for research at the intersection of climate change and mental health

The identified research priorities will only be of value if they are pursued through transdisciplinary and cross-sectoral research. The climate and mental health field is relatively new and rapidly growing, and now is the time to ensure that it is further developed and advanced to deliver a mentally healthier future in the context of the climate crisis. In a field that spans multiple disciplines and sectors, each with different cultures and ways of working, and on a topic with low awareness in many countries, what is needed to support capacity building efforts? What principles must guide the field, and what are the challenges and opportunities in the region to create an environment that would enable the realisation of the research agenda?

This section presents a synthesis – based on dialogue discussions, consultations from interviews with experts from across the region and a review of the existing knowledge base – on what is needed to implement the research agenda and foster an enabling research environment for climate change and mental health.

A desired state of climate change and mental health research in ESEA

On the topic of what *good* looks like in terms of climate change and mental health research, participants highlighted the importance of research that is culturally sensitive, locally contextualised and beneficial to the community. With the varying cultures of the communities across the regions, it is important to ensure that research efforts are adapted to the lived experiences of the communities in need. Future research at this nexus must involve the meaningful participation of communities, people with lived experiences and vulnerable groups (e.g., women, PWDs, socioeconomically disadvantaged groups, etc.); they must be treated as partners and co-creators and not mere beneficiaries and data sources. In addition, collaboration of researchers across varying disciplines and with other stakeholders was also seen to be essential in advancing this new research arena. It is important to acknowledge the complexity of the interactions between climate change, mental health and other societal factors such as socioeconomic status, history, culture, etc. Therefore, there should be diversity in the research methods used, for instance combining quantitative and qualitative methods to fully capture the nuances of climate change and mental health experiences on the ground. Additionally, scientific data and knowledge generated through research should be easily accessible and available to all stakeholders across countries and communities. Mutual exchange of knowledge, especially between the Global North and Global South, is critical; both have something novel and useful to learn from each other.

Challenges holding back research

Myriad challenges hold back research in this area, including:

- **Lack of education and awareness** of the relationship between climate change and mental health.
- **Limited funding.**
- **Limited/inaccessible data** especially on mental health outcomes, for instance coming from hospitals and primary care.
- **Limited attention on this nexus among researchers/in academia.** The few studies that are published around the subject are also not easily accessible to the non-academic community, especially among policymakers who are in charge of formulating and implementing solutions to climate change and mental health issues.
- **Diversity of cultures/contexts**, which limits the applicability of findings across the region. The predominance of studies coming from the Global North and the West makes this problem more profound when attempting to apply their research findings to Global South country contexts.
- **Lack of support** from leaders and decision makers.

Addressing these gaps is essential to generate more resources and create research environments that are conducive for climate change and mental health research.

Opportunities and enablers

Participants identified three main opportunities and enablers to achieve the desired state of research:

- **Increasing public awareness** of climate-mental health issues. The growing interest in this area has made more people join advocacy groups and demand accountability from the Global North and other major contributors to climate change while they also urge different stakeholders to find adaptation solutions through research and innovation.
- **Leveraging collaboration** between researchers across disciplines. Conferences, fora and other spaces can be used as platforms for sharing knowledge and gaining a deeper perspective on climate change and mental health research. Such collaborative spaces can also be an opportunity to gather financial, material, and human resources to support climate change and mental health research.
- **Encouraging community participation** in research, particularly with regard to the direction of research efforts during research planning and implementation.

Relevant potential partners

- Community members, particularly people with lived experience and those from vulnerable groups
- Experts from multiple disciplines, including psychology, public health, climate science, economics, business and neuroscience
- Funders, including philanthropists
- NGOs/civil society organisations
- Policymakers and government decision makers
- Private business
- Social enterprise organisations
- Universities/academic institutions

.

Priority next steps/recommendations to investors and actors

As an overall strategy to support research, participants have called to:

1. **'Decolonise' the concepts of climate change and mental health in the region**, making research frameworks, methods and approaches more culturally-appropriate to the needs and experiences of our communities.
2. **Use clear language that is accessible to the communities and vulnerable groups** who will act as research partners; scientific inquiry should be understandable to all.
3. **Create capacity building opportunities for current and future researchers** to build an academic community that values leadership and growth; this is critical for sustaining climate change and mental health research in the long term. This strategy includes incorporating climate change and mental health into the research agendas of educational institutions, including departments of various disciplines, to help increase awareness of these issues and encourage more collaboration for research.
4. **Advocate for greater diversity in methods and research designs** in climate change and mental health research.

Translating a growing evidence base into action that can respond to the mental health impacts of climate change

With the urgency and scale of the intersecting climate change and mental health crisis, action at all levels is necessary to address escalating mental health needs and promote co-beneficial climate actions. Therefore, current and future evidence base on the interconnections between climate change and mental health must drive and inform action in the arena of policy and practice. How can current evidence and new insights generated through research be best translated into changes to policy decisions and practices across both climate and mental health regional spaces?

This section presents a synthesis of dialogue discussions and pre-dialogue scoping, setting out the challenges and opportunities in relation to translating evidence generated through research into policy and practice.

A desired state of climate change and mental health to action in ESEA

Grounded research leads to grounded policies, practices and interventions that are locally tailored and culturally appropriate, based on the lived experiences of communities. Climate change and mental health actions must be contextualised to local community needs and characteristics, incorporating their traditional knowledge and spoken languages. These actions and policies must demonstrate inclusivity and transparency while ensuring the protection of the rights of vulnerable groups and communities and the promotion of their interests. Ultimately, future climate change and mental health policies must be founded on climate justice, ensuring that major contributors to climate change are held accountable and communities that are disproportionately impacted receive utmost care. Similar to the research arena, there should also be genuine collaboration between various stakeholders across sectors and disciplines in the formulation, implementation, and evaluation of policies and interventions. Such transdisciplinary and cross-sectoral collaboration is vital for the sustainability of actions and results. Communities must also be involved in the monitoring and evaluation of policies and interventions; timely feedback from people in communities can inform the continuous improvement and iteration of solutions. Climate

change and mental health policies and interventions must be dynamic, adaptive and forward-looking, especially in anticipation of the future effects of climate change.

Challenges holding back translation of evidence

CCM dialogues surfaced numerous challenges preventing the effective translation of evidence to action, including:

- **Lack of awareness** among policymakers and governments on climate-mental health issues, **which hinders action**. Some policymakers and practitioners may not be well-versed in research and therefore face difficulty in understanding scientific data without the help of academics.
- **Limited/inaccessible data on climate change and mental health**; this data is also rarely integrated, making comprehensive analysis difficult.
- **Limited funding and prioritisation** of climate-mental health issues. The intersection of these fields is a new area of inquiry, and therefore research money for this subject is almost non-existent in the region.
- **Politicisation** of climate-mental health issues, which can hinder the development of climate change and mental health policies, programs, and funding initiatives; this is especially true when conservative politicians deny the urgency of climate change, ignore the mental health crisis, and villainise climate change and mental health activists.
- **Diversity of cultures/contexts** preventing inter-country collaboration and the development of shared universal policy agendas.
- **Limited collaboration/silos** across relevant sectors.

The hope is that this initiative will help break the silos and foster greater collaboration for climate change and mental health, not only in research and advocacy but also in policy and practice.

Opportunities and enablers

Opportunities and enablers to translate evidence to policy and practice include:

- **Increasing engagement** between researchers, government leaders, policymakers and implementers, particularly through joint discussions, forums and conferences.
- Deep, respectful **collaboration with communities**. Such collaboration must go beyond awareness raising and encompass intervention design and implementation as well as monitoring and evaluation.
- **Capitalising on the increasing awareness among government leaders** of climate-mental health issues. We should use this as an opportunity to grow support for climate change and mental health policies and programs and demand accountability from major climate change contributors.

Relevant potential partners

- Community members, particularly people with lived experience and those from vulnerable groups
- Government policymakers, decision makers and implementers involved in climate change (e.g., ministries of environment, climate change policy bodies) and mental health (e.g., ministries of health, national public health institutes)
- Media/journalists
- NGOs/civil society organisations
- Universities/academic institutions

Priority next steps/recommendations to investors and actors

In order to effectively translate evidence into action, we need to:

1. **Strengthen the links between researchers and policymakers** by creating avenues where they can openly discuss issues and develop solutions to climate change and mental health problems.
2. **Empower communities and increase public awareness**, especially among professionals and policymakers, on climate change and mental health to encourage open discussion and generate support for these issues.
3. **Evaluate the effectiveness of current policies** to address the needs of populations in terms of climate change, mental health and their intersection.
4. **Conduct training and capacity building workshops for both researchers and policymakers** to improve communication of evidence and solutions between these groups and to broader audiences.

Discussion: strengths, limitations and next steps for the research and action agenda

Strengths

This regional research and action agenda has several strengths that can significantly advance the scientific and policy discourses on the climate change and mental health intersection. Experts who participated in the dialogues have technical expertise in fields relevant to climate change and mental health as well as on-the-ground lived experiences of climate change, since ESEA is one of the most climate-vulnerable regions globally. The insights generated from the dialogues represent the diverse cultures in the region, which ensures the incorporation of nuanced traditional knowledge and solutions to the regional response to climate change and mental health. Additionally, this document surfaces needs and experiences that were not previously heard or captured in literature, as articulated by representatives from various vulnerable and marginalised groups. The dialogues also significantly contributed to raising awareness and visibility of the nexus of climate and mental health among diverse stakeholders in the region.

Limitations

However, the agenda is not without limitations. We struggled to balance the urgency to develop a research and action agenda for this rapidly growing field with the importance of meaningful convening and relationship building, which cannot be rushed and must move at the pace of trust. Striking this balance is especially important when engaging with groups frequently subjected to extractive research processes (i.e., Indigenous Peoples, disaster survivors, etc.). The region is also diverse in terms of geography (separated by thousands of islands and territories), culture and languages (hundreds of cultures and dialects). Future projects face the challenges of balancing the need for standardisation with local contextualisation, including the use of local terminologies and incorporation of cultural understandings. Nonetheless, these limitations can also catalyse the development of localised methods and metrics to further understand people's experiences and reactions in relation to climate change and

mental health. Despite our best efforts, certain groups were not engaged throughout the process; for instance, communities that do not speak English, which is the region's common language. Future research and advocacy activities must therefore attempt to reach these populations.

Next steps

The next major challenge is how to sustain the connections built in this new regional community of practice and ensure that this regional research and action agenda comes into fruition.

Conclusion

A promising future lies ahead for the emerging climate change and mental health community in ESEA, especially in terms of research that is collaborative, innovative and responsive to people's needs. Our rigorous process comprising pre-dialogue scoping, regional dialogues and expert interviews produced a better understanding of the landscape of climate change and mental health research in the region. There is rapidly growing shared interest among academic experts, policymakers, private sector, civil society and young people in pursuing joint scientific and advocacy efforts to promote further understanding of, and generate solutions for, the mental health impacts of climate change. The stakeholders involved in this process shared the same sentiment: our regional response to climate change and mental health must be contextualised and community-driven, taking into account the specific and unique characteristics and needs of populations and groups in the region, especially among those coming from vulnerable backgrounds. The existing scholarship as well as initiatives in policy, education and advocacy provide a glimpse of what is yet to come for the climate change and mental health field in the region.

Research themes and questions generated from the different activities of this process highlighted the urgency of addressing the climate crisis and its mental health consequences to the countries and communities in the region, especially among those populations that are most vulnerable to climate hazards and mental health disorders (e.g., children, the elderly, low-income groups, farmers, fisher peoples, people with disabilities and existing health issues, etc.). Several proposed potential pathways and mechanisms – such as socioeconomic inequities, gender-based violence, social stigma, spiritual connection, etc. – showcase the intricate and multi-faceted links between the climate crisis and mental health in the region. Research themes also covered climate adaptation and mitigation measures as potential sources of mental health benefits, and existing and potential mental health interventions that can protect mental health from climate impacts.

Overall, our vision is an ESEA where people live and thrive, not only physically but also mentally and emotionally, in spite of the climate crisis, while also harnessing their personal and community resilience to drive action to tackle the climate crisis. To achieve this vision, we need collaborative research that is transdisciplinary, cross-sectoral, international and multigenerational; it must be anchored in equity, justice and a deep commitment to applying research findings and lessons learned to inform policies and practices that will promote and protect the region's collective mental health in the era of a warming planet.

Hearing from the Regional Community of Practice

“The young people, who face the brunt of climate change, are the most vocal in calling out for ambitious climate action, yet they are seldom supported with their climate-related endeavours.” - Jasmin Irisha Jim Ilham, Youth Ambassador, during Dialogue 1

“The term ‘mental health’ is not necessarily something that communities use on an everyday basis. There’s a need to localise mental health and dimensionalise it in culturally relevant ways.” - Dialogue 1 participant

“Most of the data is still oriented towards mental illness. How do we move towards collecting data for mental health wellbeing?” - Dialogue 1 participant

“Different groups of people are differently vulnerable to the same climate change impact. For example, it’s not necessarily the case that women are more vulnerable to climate change than men. Women can be more resilient to climate change than men in some respects, and men can be more vulnerable to climate change than women in other respects. It’s less about identifying a more vulnerable group but identifying the factors that make people become differently vulnerable to certain climate change factors.” - Dialogue 1 participant

“I will continue sharing my experiences facing adverse climate events with the hope of preventing others from going through the same mental and moral distress, knowing we are not alone in this.” - Dr. Madihah Ahmad Puaad, Lived Experience Advisory Group member, during Dialogue 2

“I am glad to have the opportunity to contribute to the Connecting Climate Minds project. I am more aware of the urgent needs for more research and support for a growing population whose livelihoods and mental health is impacted by the changing climate. As a researcher myself I am more eager to conduct more research in this area and work more on ways I can do my part to mitigate climate change in my daily life. Thank you to everyone in the CCM team who brought this project to life.” - Niaya Harper Igarashi, Regional Co-convenor (HGPI)

Glossary of terms

For a glossary describing relevant concepts and key words for the Connecting Climate Minds research and action agendas, please download from [here](#).

Who produced this report

Author contribution statement

This development of this report and the conduct of this project were led by the Planetary and Global Health Program (PGHP) of the St. Luke's Medical Center College of Medicine - William H. Quasha Memorial in the Philippines - the Eastern and South-Eastern Asia regional community convener (RCC) of Connecting Climate Minds. This report was written under the overall direction and leadership of Renzo R. Guinto, director of the PGHP and lead of the Eastern and South-Eastern Asia RCC. John Jamir Benzon R. Aruta led the analysis of dialogue outcomes, pre-dialogue scoping, and drafting of major sections of this report, with support from Kent Tristan Esteban and Jun Angelo Sunglao. Gail S. Co coordinated the different activities of the project - especially the two dialogues - that generated findings for the report. Dianne J. Cunanan and Alfredo Lorenzo R. Sablay both supported the implementation of the dialogues. All team members of the PGHP contributed different portions of the report and participated in various aspects of the dialogue as organisers and facilitators. Colleagues from one of the co-conveners, Health and Global Policy Institute in Japan, particularly Joji Sugawara and Niaya Harper Igarashi, also participated in the dialogue as facilitators and contributed inputs to the writing of this report. Emma L Lawrance was the Principal Investigator for the Connecting Climate Minds project, worked on project conceptualisation and supervision and contributed to editing this report. Jessica Newberry Le Vay contributed to conceptualisation and data management. Nienke Meinsma was the project manager for Connecting Climate Minds. Peter Howitt was the Eastern and South-Eastern Asia Regional Community supporter, providing on-going support to the RCC from the global team. Didi Thompson copyedited the report.

This is the independent work of the authors with the support and input of the Connecting Climate Minds team.

Acknowledgements

The content of this report has been produced through a collaborative process with various experts and stakeholders across Eastern and South-Eastern Asia. We acknowledge the invaluable contributions of those who shared their expertise, voices, and ideas in shaping this research and action agenda. Accountability for the content of this report lies with the authors, but we share ownership of this work with all these contributors. We would like to acknowledge the contributions made by (in alphabetical order, along with their contributions):

Shaine Therese Aguado (Dialogue 2)

Julius Cesar Alejandre (Dialogue 1, Dialogue 2)

Suhailah Ali (Dialogue 1 Notetaker)

Rowalt Alibudbud (Dialogue 1, Dialogue 2, Post-Dialogue Survey)

Ashwena Apao (Dialogue 1, Pre-Dialogue Survey, Post-Dialogue Survey)

Johannes Reiner Asio (Dialogue 1 Notetaker)

Louella Abigail Asuncion (Dialogue 1 and Dialogue 2 Notetaker)

Emmanuel Benyeogor (Pre-Dialogue Survey)

Marianne Bongcac (Dialogue 1 and Dialogue 2 Notetaker)

Veronica Cabe (Dialogue 1, Dialogue 2, Pre-Dialogue Survey, Lived Experience Advisory Group)

Dr. Pamela Cajilig (Dialogue 1, Dialogue 2, Pre-Dialogue Survey)

Dr. Timothy Hudson David Carandang (Dialogue 1 Notetaker)

Dr. Katrina Ceballos (Dialogue 1 and Dialogue 2 Facilitator)

Dr. Kai-Yuan Cheng (Dialogue 1, Dialogue 2, Pre-Dialogue Survey)

Dr. Gopalasamy Reuben Clements (Dialogue 1, Pre-Dialogue Survey)

Dr. Jake Bryan Cortez (Dialogue 1 Facilitator)

Charlie Choy (Dialogue 1)

Dr. Yel Daravuth (Dialogue 1, Pre-Dialogue Survey)

Silvina Idalina de Jesus (Post-Dialogue Survey)

Laetania Djandam (Dialogue 1 Facilitator, Post-Dialogue Survey, Regional Co-convenor)

Laily Lutfiana Dhia (Dialogue 1, Dialogue 2, Pre-Dialogue Survey, Post-Dialogue Survey, Lived Experience Advisory Group)

Elisa Ximenes dos Santos (Post-Dialogue Survey)

Dr. Kimin Eom (Dialogue 2, Expert Interview)

Maricel Estavillo (Dialogue 2)

Dr. Crystal Estrada (Dialogue 2)

Dr. Gabriela Fernando (Pre-Dialogue Survey)

Caroline Fried (Dialogue 1, Dialogue 2)

Rodne Galicha (Dialogue 1)

Dr. Johnrev Guilaran (Expert Interview)

Dr. Sharina Abdul Halim (Dialogue 1, Dialogue 2)

Iman Fahim Hameed (Dialogue 1, Dialogue 2, Pre-Dialogue Survey)

Dr. Tran Thi Tuyet Hanh (Dialogue 1, Pre-Dialogue Survey)

Charlotte Hicks (Dialogue 1, Post-Dialogue Survey)

Pengsan Huon (Pre-Dialogue Survey)

Yuka Iijima (Dialogue 1, Dialogue 2, Post-Dialogue Survey)

Jasmin Irisha Jim Ilham (Dialogue 1 Facilitator, Youth Ambassador, Post-Dialogue Survey)

Marieanne Itol (Dialogue 1, Dialogue 2, Pre-Dialogue Survey)

Jeta Jiranuntarat (Dialogue 2)

Maria Puspa Kartika (Dialogue 1 Facilitator, Dialogue 2, Pre-Dialogue Survey, Post-Dialogue Survey, Regional Co-convenor)

Audrey Kianjaya (Post-Dialogue Survey)

Dr. Yoonhee Kim (Pre-Dialogue Survey, Expert Interview)

Dr. Ivan Kurnia (Dialogue 1, Pre-Dialogue Survey)

Dr. Honey Libertine Achanzar-Labor (Dialogue 2, Expert Interview)

Alyanna Lagasca (Dialogue 1)

Michelle Lapiz (Dialogue 2 Notetaker)

Aishath Thimna Latheef (Dialogue 1, Dialogue 2, Pre-Dialogue Survey, Post-Dialogue Survey)

Lincoln Lau (Post-Dialogue Survey)

Elaine Clarisse Lim (Dialogue 1, Dialogue 2)

Mathea Melissa Lim (Dialogue 1, Dialogue 2)

Trisha Lim (Pre-Dialogue Survey)

Dr. Mingnan Lin (Dialogue 1, Dialogue 2, Expert Interview)

Syaru Shirley Lin (Dialogue 2)

Dr. Muhammad Daniel Azlan Mahadzir (Dialogue 1, Dialogue 2, Pre-Dialogue Survey, Lived Experience Advisory Group)

Dr. Alberto Teodoro “TJ” Malvar (Pre-Dialogue Survey)

Oyungerel Munkhbat (Dialogue 1)

Prof. Keiko Nakamura (Dialogue 1, Connecting Climate Minds Regional Advisory Board)

Kirchelle Ann Mae Nodado (Dialogue 2 Notetaker)

Sheen Orihuela (Dialogue 1 Notetaker)

Adi Palguna (Dialogue 1, Dialogue 2, Pre-Dialogue Survey, Post-Dialogue Survey)

Joonha Park (Dialogue 2)

Shervin Pascual (Dialogue 2 Notetaker)

Jhonatan Yuditya Pratama (Dialogue 1, Dialogue 2, Pre-Dialogue Survey, Lived Experience Advisory Group)

Dr. Madihah Ahmad Puaad (Dialogue 1, Dialogue 2, Pre-Dialogue Survey, Post-Dialogue Survey, Lived Experience Advisory Group)

Dr. Luis Carlos Rodriguez (Dialogue 1, Pre-Dialogue Survey)

Dr. Connie Cai Ru (Dialogue 1, Dialogue 2, Pre-Dialogue Survey)

Sena Salcedo (Dialogue 1, Dialogue 2, Psychosocial Support Adviser)

Dr. Moises Neil Serino (Pre-Dialogue Survey)

Saumya Surbhi (Pre-Dialogue Survey)

Sarah Queblatin (Dialogue 1, Dialogue 2)

Mitzi Tan (Dialogue 2)

Midori Tatsumi (Pre-Dialogue Survey)

Dr. Pauline Tiangco (Dialogue 2 Facilitator)

Bao To (Pre-Dialogue Survey)

Lien To Thi (Dialogue 1, Dialogue 2, Pre-Dialogue Survey, Post-Dialogue Survey, Regional Co-convenor)

Angela Tritto (Dialogue 1, Dialogue 2, Pre-Dialogue Survey, Post-Dialogue Survey)

Naw Aye Chan Wadi (Dialogue 1, Post-Dialogue Survey)

Naw Chit Wai Wai (Pre-Dialogue Survey)

Dr. Grace Wangge (Dialogue 1, Pre-Dialogue Survey)

Johannah Wegerdt (Pre-Dialogue Survey)

Wang Shiou Wu (Post-Dialogue Survey)

Shakila Zen (Dialogue 1, Dialogue 2, Pre-Dialogue Survey)

Yves Zuniga (Dialogue 1)

Funding

Funding for this work was provided by Wellcome.

Conflicts of interest

The authors have no conflicts of interest to declare.

Appendices

Appendix 1: Dialogue agendas

Dialogue 1

Time (mins)	What	How
0-20	Welcome and introductions	
20-35	Visioning Exercise	Visioning Exercise: “Headlines” on Climate Centre Good Games platform Participants share what they are hoping to get from being part of Connecting Climate Minds, and why they are invested in this topic <i>Imagine we are in 2030 and the CCM project has received an important prize for its achievements in bringing CC/MH together in your region and making change for climate and mental health together through research and action. You are interviewed by a journalist as you are celebrating winning the prize.</i> Create / vote on and review Headlines created on the Good Games Platform
35-50	Scene setting	Brief context to the links between climate change and mental health What are we hearing in the region that we can come together and respond to: sharing highlights of pre-scoping Presentation on Connecting Climate Minds and Q&A
50-95	Breakout session 1: Regional needs	Host sets scene for breakout discussion Share climate hazards overview for the region Breakout groups

		Facilitator to lead the group in a discussion on the mental health challenges arising from climate change in the region. Questions posed to participants were: • <i>From your perspective, what are the climate hazards that concern you the most in terms of mental health and wellbeing?</i> • <i>For each hazard, discuss the following:</i> ○ <i>Who is being impacted? Are there any groups more at risk?</i> ○ <i>What are the mental health and wellbeing outcomes associated with this hazard?</i> ○ <i>How does this hazard specifically affect mental health and wellbeing; what are the mechanism e.g. housing, livelihoods etc</i> ○ <i>Encourage people to look at different types of mechanisms e.g. social, biological, cultural, environmental</i> ○ <i>Climate adaptation/mitigation actions - what is being done, what can be done?</i> ○ <i>Actions that are being/could be taken to improve or promote mental health and wellbeing in this context</i> Brief plenary highlights
95-105	Break	
105-110	Sneak preview	The cartoon artist shares one first draft reflecting some of the content of the session so far. Host invites participants to share in the chat how the image resonates.
110-160	Breakout session 2: Evidence gaps and research priorities	Breakout groups [same participants, facilitator and note-taker as first breakout session] Looking back at the hazard table from Breakout session 1 as a starting point, participants are guided to start thinking about where more research is needed. Hosts invite 2 or 3 facilitators to share some brief reflections
160-170	Cartoon reflection on dialogue output	Participants individually review each cartoon, and share their reflections. This can range from: suggestions for changes to make it more accurate, to sharing how it relates to our reality, or any other insights.
170-180	Wrap-up and next steps	Host shares a high level recap of the session, thanks everyone and celebrates this first coming together of the regional community.

Participants complete brief survey to provide feedback on the dialogue

Dialogue 2

Time (min s)	What	How
0-5	World Map Exercise	Host asks participants to place a “star” icon where they currently are and then take the “heart” icon and place it where they would like to visit one day or a place they have been and really loved/feel especially connected to.
5-15	Welcome and introductions	
15-30	Sharing updates	Presentation on global and regional progress and outcomes of dialogue 1, including vision, priority research themes.
30-50	Priority Research Theme Feedback	Jamboard shared on screen. Host introduces priority research theme activity and asks people to consider the following questions: Research questions ● <i>Is this a valuable theme? Would you add or change anything?</i> ● <i>Share any ideas on datasets, metrics or methods that could be applied to help address this.</i> Research topics ● <i>What would be most helpful to know about this topic to understand and respond to it?</i> ● <i>What would you want to know first?</i> Music plays and participants work independently on Jamboard at their own speed.

50-55	Scene setting for breakouts	Host sets the scene for the breakout discussion. Interested in the 'how', which includes the principles and desired outcomes for research, policy and practice in the region. The discussion will enable targeting of action and investment to both implement the research agenda in the region, and to ensure evidence on the climate and mental health nexus is translated into policy and practice
55- 90	Break out 1	Eight breakout groups. The first 4 on the theme <i>“Creating knowledge through research”</i> (e.g. How can we best implement the research agenda?) and the other 4 on <i>“Fostering evidence-based policy and action”</i> (e.g. How can we translate the growing evidence base to action in policy and practice?). Each group uses a Jamboard to discuss: 1) What does this look like when it is done well? 2) What opportunities & enablers exist? 3) What challenges must we overcome? 4) Who should our partners and stakeholders be? Brief plenary highlights
90-100	Break	
100-105	Draw your feelings	Host welcomes everyone back and invites all who'd like to join to grab a piece of paper and a pen/marker (or digital) - and draw how they are feeling today. Anyone willing is invited to show their image - share some reflections.

105-125	Spectrum Mapping	Host introduces and explains the activity. Share screen of Jamboard. Designed to reveal the diversity of perspectives and options around different concepts and to organize them into a meaningful spectrum. This activity aims to illuminate the group's range of perspectives and to organize those perspectives into a continuum. Question: <i>What do these concepts mean to you?</i> Focusing on one board at a time, host asks everyone to silently generate a point-of-view around that concept and write it on a sticky note on the Jamboard. People can add more than one. Once the sticky notes are posted, the host works with the group to sort them into a horizontal range of ideas. Sticky notes that express similar perspectives or options should go next to each other. Sticky notes that seem to be outliers should stand alone; they may sometimes end up defining the limits of the range. Repeat with the remaining concepts. At the end, ask for observations and insights on the lay of the land. And if any perspective or option has been excluded. If so, add it and re-sort as necessary.
125 - 155	Break out 2	Host introduces the second round of breakout rooms and provides a 2-minute summary of the Jamboard contributions from the first breakout. Original 8 groups reconvene and expand and add to their initial thoughts. Groups are allowed to continue on the theme they were already discussing or switch to the other theme.
155- 175	I will ... I want...	Host asks everyone to personally reflect about how they will contribute to this space, and share an "I will" statement in the chat, but only press enter when told to do so. After a minute, host invites everyone to press enter at the same time to make statements appear. This is repeated with a statement about "Out of this community I would like to see..."

175	Wrap up and next steps	Host shares a high level recap of the session, thanks everyone and outlines next steps. Participants complete brief survey to provide feedback on the dialogue
-----	-------------------------------	--

Appendix 2: Structure and selection criteria for priority research themes

Structure of the priority research themes:

They will nearly always need to include:

1. Population
2. Outcome [relevant mental health challenge(s)]
3. Exposure and/or intervention [climate hazard, climate action or mental health action]

May be appropriate to include:

1. Pathways or mechanism being studied
2. Factors conferring vulnerability or resilience that are particularly of interest or relevance to the theme
3. A comparison (e.g. another intervention or no intervention)
4. Timeframe: over what timescale would this research be conducted or applied? Timescale of the climate hazards it is targeting? Timescale over measuring mental health outcomes (e.g. is it immediate impacts during a heatwave or longer term post traumatic stress many years after a flood?)
5. Geography/scale: is it applicable across the whole region, or is relevant to specific locations (e.g. one localised geography, urban heat islands)
6. Level: is this applicable to individuals, families, communities or systems?

Examples could include:

- What is the prevalence of suicidal ideation and deaths by suicide in farming communities experiencing drought in Sub-Saharan Africa?
- Does use of psychotropic medications affect people's ability to regulate body temperature during periods of extreme heat?
- What are the benefits to mental health and emissions reductions of increasing green space in low income urban areas?
- What are the benefits to mental health from training community-based healthcare providers to provide mental health support following typhoons in the Philippines?

Selection Criteria:

Selection criteria (should be objective and standard across regions. Drawn from workshop discussion and [NSW Government guide to setting research priorities](#))

- Does the research theme answer an existing evidence gap? Cross checking with pre-dialogue scoping, with evidence gaps data from dialogue
- Does the research theme have potential to inform decision making in policy and practice?
- Does it reflect one or more of the following:
 - the identified greatest mental health challenges linked to climate change in the region (cross check with pre-dialogue scoping, pre dialogue survey., breakout 1 data)
 - identified greatest climate hazards in the region
 - evidence policymakers most need to address emerging needs / overcome barriers to action (from breakout 1 data)
 - Lived experience needs identified (Does it align with what the experts by lived experience are saying are challenges in the region?)
- Could it be feasible to undertake this research?
- Is it framed to be understandable, specific and practical for the research community to meaningfully address? Could a researcher or a funder with relevant expertise understand this theme?
- Is it specific enough to be meaningful and guide focussed research (if too broad or vague it won't help target resources meaningfully)
- It helps funders invest in this area (consider what is regionally relevant here)
- *Is it possible/useful to include a cost component here? NSW Govt guidance includes a suggestion on anticipated return on investment through use of evidence in policies, services or programmes*

Relevant information to capture and consider for each theme

- Who priority was raised by i.e. prioritising themes that have been raised by a diversity of voices (i.e. expert researcher and lived experience expert and practitioner)
- Potential to address health inequities / does it target a population particularly vulnerable to MH impacts of climate change
- Can existing datasets/metrics/indicators be used? (cross-checked with mid-dialogue survey / dialogue 2 data). It is fine to include themes where this isn't yet true - we will start to explore how datasets can be expanded, and methods combined across disciplines in dialogue 2.
- Do required research capacities exist in the region? I.e. this building on existing research strengths or is this a longer term priority requiring capacity building
- Relevance to current policy context / tied to anticipated window of opportunity (Cross checking with pre-dialogue scoping, with evidence gaps data from dialogue)
- Scale and time frame e.g. could have a really pressing question that only apply to small geographical region. Scale in terms of individual, community or systems level.

Appendix 3: Research agenda coding framework

Research Category	Sub-Category	Example Code
1. Impacts, Risks, and Vulnerable Groups This category is about improving our understanding of the ways in which mental health is affected by climate change. This category also includes the ways we can go about getting this improved understanding of mental health impacts of climate change - the methods and metrics we need to assess and monitor mental health in ways that are relevant to climate change, contextually appropriate, comparable etc.	1.1 Prevalence and nature of mental health outcomes	1.1 Anxiety and trauma from experiencing climate hazards
	1.2 Quantifying the mental health burden	1.2 Number of people affected by depression due to disasters
	1.3 Risk factors	1.3 Geographical location as a risk factor
	1.4 Vulnerable sub-groups	1.4 Farmer suicides due to droughts
	1.5 Quantifying the cost burden	1.5 Challenge in calculating loss and damage on mental health
	1.6 Methods research to assess and monitor mental health impacts	1.6 Introducing a psychosocial support tool to quantify the mental stress of the community
2. Pathways and mechanisms This category is about improving our understanding of how mental health is affected by climate change. We are interested in research themes that can help identify, categorise and understand the range of ways that climate change or climate action may act to affect mental health. This can include considering pathways and mechanisms that are biological, psychological, societal or environmental in nature, and may be direct or indirect.	2.1 Societal Mechanisms	2.1 Flooding causes inaccessibility to basic needs leading to PTSD
	2.2 Environmental Mechanisms	2.2 Acute climate change events cause direct mental health outcomes such as trauma
	2.3 Psychological Mechanisms	2.3 Climate change exacerbating symptoms of those with preexisting mental health issues
	2.4 Biological Mechanisms	2.4 Air pollution causing inflammation of brain cells and increasing risk to psychosis
	2.5 Mechanisms for Co-Benefits	2.4 Pathways and mechanisms of nature-based solutions on mental health
	2.6 Methods research to assess and monitor pathways and mechanisms	2.6 Qualitative studies

3. Mental health benefits of climate action [adaptation and mitigation] This category is about how climate adaptation and mitigation actions, across sectors, can also have win-win benefits for mental health. This includes quantifying costs and benefits of climate action for mental health, understanding what is needed to support better alignment between climate action and mental health action, and identifying where this integration is already happening across strategies and policies.	3.1 Climate actions with mental health benefits	3.1 Integration of green spaces to improve mental health
	3.2 Quantifying co-benefits	3.2 What specific climate actions contribute to mental health?
	3.3 Differentiation of costs and benefits by population sub-groups	3.3 Comparison across socioeconomic status
	3.4 Government structures and decision support tools to enable action	3.4 Strengthen local health systems for climate change and mental health interventions
	3.5 Integration of mental health into adaptation and mitigation policies	3.5 Mandate mental health integration in emergency or disaster response
	3.6 Integration of mental health into other climate priority areas	3.6 Climate financing in adaptation and mitigation to enhance the distribution of social protection
	3.7 Climate Action within the mental health sector	3.7 Adapting the health system to be more climate resilient and environmentally sustainable
	3.8 Methods research to assess and monitor mental health benefits of climate action	3.8 Is climate change awareness resulting in positive or negative mental health and behavior?
4. Mental health interventions/solutions in the context of climate change This category is about identifying the most effective mental health interventions/solutions to support mental health in the context of climate change. This might be about providing support to people already experiencing negative mental health impacts, or about reducing risk or severity of	4.1 Ways of knowing that build resilience	4.1 Traditional practices to help build resilience after typhoons
	4.2 How existing mental health interventions are affected by climate change	4.2 Damaged of infrastructures due to climate events limiting hospital access
	4.3 Evaluating mental health interventions that integrate climate	4.3 Forest bathing among school children to increase appreciation of nature and improving mental health
	4.4 Adapting mental health interventions for climate-related impacts	4.4 Counseling and trauma debriefing for disaster responders

future negative mental health impacts. This includes learning from knowledge held by different disciplines, communities and cultures, understanding how existing mental health interventions are affected by climate change, identifying and evaluating existing interventions that are relevant to the context of climate change, and developing new interventions. Interventions are relevant at all levels (individual, family, community, systems) and across sectors.	4.5 Novel interventions to meet climate-related mental health needs	4.5 Establishing a community of practice for climate change and mental health as a possible intervention
	4.6 Determining 'best buys' (intervention effectiveness across settings and subgroups)	4.6 Best practices of climate change and mental health services and programs between rural and urban communities
	4.7 Workforce capacity building to meet mental health needs of the climate crisis	4.7 Training of mental health professionals for on-the-ground support

Appendix 4: Action agenda coding framework

Research Category	Sub-Category	Example Code
1. Creating an enabling environment for research at the intersection of climate change and mental health	1.1 Desired state of research	1.1 Meaningful engagement with vulnerable groups
	1.2 Opportunities and enablers	1. 2 Population is becoming more aware of climate change and mental health
	1.3 Challenges holding back research	1.3 Western frameworks that are not culturally appropriate are used
	1.4 Partners/stakeholders	1.4 Learning and research institutions
	1.5 Priority next steps/recommendations to investors and actors	1.5 Capacity building for Global South researchers
2. Translating a growing evidence base into action that can respond to the mental health impacts of climate change	2.1 Desired state of evidence to action in policy and practice	2.1 Contextualised policies grounded on the real-world experiences
	2.2 Opportunities and enablers	2.2 Some leaders are recognising climate change and mental health
	2.3 Challenges holding back action	2.3 Some researchers cannot communicate research findings effectively
	2.4 Partners/stakeholders	2.4 Government and policymakers

	2.5 Priority next steps/recommendations to investors and actors	2.4 Collaboration with learning institutions on conducting workshops for communicating research data to policymakers
--	---	--

Appendix 5: Data collection and storage

Collecting data

- **Informed consent** was obtained before any data is collected. Consent processes are outlined for specific project activities below.
- All **participants must be aged 18** or over to comply with our ethics approval.

Storing data

Climate Cares was responsible for storing all project data on a secure server at Imperial.

Imperial does not have a collaborative secure research environment that is accessible by non-Imperial users, so all project partners who collected, received and analysed data were responsible for securely storing and processing data in line with their own organisation's policies.

This includes responsibility for:

- Storing data securely in line with own organisation's policies
- Transferring data files securely
- Deleting files if no longer needed
- Having appropriate data sharing agreements in place with partners also involved in data analysis (e.g. RCC Co-Convenors and/or experts involved in project outputs)

Sharing data

Receiving data from Climate Cares

Climate Cares shared data with partners (e.g. survey data) via Imperial College's secure data sharing process (Transfer).

Sending data to Climate Cares

Ways of sharing data with Climate Cares will be either using the same method as above (Transfer), or through Qualtrics.

Sending data to other partners

Data shared with other partners (e.g. RCCs sharing data with Co-Convenors for analysis) was done in whatever ways work best for the organisation. Appropriate data sharing agreements in place were considered.

Data processes for project activities

Surveys

- Data collection: Qualtrics, collected by Climate Cares
- Informed consent: captured via Qualtrics and stored by Climate Cares

- Participants were asked if they wanted to consent to being potentially contacted to share their experience of being involved in the project, or lived experience story. Names of participants who said yes were shared with relevant project members.
- Data sharing:
 - Use of survey for dialogue sign ups: Climate Cares shared names of people who have said they will attend the dialogues, any language and accessibility needs, with RCCs and Lived Experience Working Group via the relevant invite spreadsheets
 - Analysis of survey data: Climate Cares shared anonymised survey data with relevant project members (RCCs and Lived Experience Working Group) for analysis within their own groups.

Dialogues

- Data collection: Zoom platform was used to conduct the dialogues. Climate Centre saved the file for the main session Zoom recording while break out facilitators saved files for their respective break outs. Notes from dialogues were captured during the dialogue in shared Google Docs.
- Informed consent was collected via Qualtrics in pre-dialogue survey.
- Data sharing:
 - Zoom recordings
 - Instructions for what to do with Zoom recording files were as follows:
 - Facilitators / notetakers
 - Record breakout sessions and save to their local computer
 - Send recordings to Climate Cares via secure Transfer link that will be emailed to them in advance
 - Receive confirmation from Climate Cares that recordings have been received
 - Delete the recordings from their local computer
 - Sign form to confirm they have deleted the recordings
 - Climate Centre stored plenary session recording and sent them to Climate Cares via Transfer.
 - Notes
 - Immediately after the dialogue, the collaborative Google Doc version of the notes was moved from Google Docs to be securely stored at Imperial and shared back (immediately) as offline copies with RCCs via Transfer.
 - Transcription
 - Climate Cares worked with a transcription service (Way With Words) to transcribe dialogue discussions. Anonymised transcripts was shared with relevant project members (e.g. RCCs) via Transfer.
 - Analysis
 - Where project members are sharing data outside of their own organisations (e.g. RCCs sharing with analyst teams), they are responsible for ensuring secure data sharing.

Appendix 6: Climate hazards in Eastern and South-Eastern Asia ³²

Observed climate-related hazards in ESEA and projected changes

To guide informed discussions on the current and potential mental health consequences of current and future climate change in the region, it was vital to ensure a grounding in climate science. The Red Cross Red Crescent Climate Centre (RCCC) conducted a mapping of previous and projected climate hazards across the region to inform the dialogues and research agenda. The following is a summary of the climate-related hazards that Eastern and South-Eastern Asia has experienced over the last 30 years, and their increases projected to approximately 2030.

It is important to emphasise that the data gathered for the past and future periods differ not only in spatial resolution but are also based on different underlying principles. The former are through observations or documented occurrences such as those reported by national weather services and are aggregated at a country-scale. The latter are from climate modelling-based studies and assessment reports published by the Intergovernmental Panel on Climate Change (IPCC), which are at a more granular spatial scale. For the same reason, the categories of climate-related hazards in the past and future periods may at times not be identical but are generally comparable or related. For instance, flooding that is reported as a relevant climate hazard in the last 30 years here is not a direct output of global climate models (GCMs). Instead, the underlying meteorological variable (rainfall) from the simulations of GCMs are used for deriving indices that can be considered as a proxy for potential flooding in a future warming climate.

ESEA

The Eastern and South-Eastern Asian region is one of the most vulnerable to climate change. This region regularly experiences strong droughts, wildfires, floods, landslides, storms and extreme temperatures over the recent years, as international disaster database EM-DAT (<https://www.emdat.be/>) sources make evident. Note that the limited number of extreme events recorded (e.g. droughts and wildfires) may be due to the missing records in the EM-DAT data for various reasons, such as poor or limited network of satellite and surface observations over the region, or limited reporting from national meteorological services. In addition, the spatial aggregation of the data to country-scales makes it difficult to map the occurrence of the past hazards to human settlements. The total number of people affected as reported in EM-DAT therefore may not always be truly representative of the frequency of occurrences of the underlying climate hazard. For instance, a country in Eastern and South-Eastern Asia may have experienced frequent heatwaves, but if these events were not recorded or in uninhabited regions, the total number of occurrences discussed below may be an underreporting.

Previous Droughts

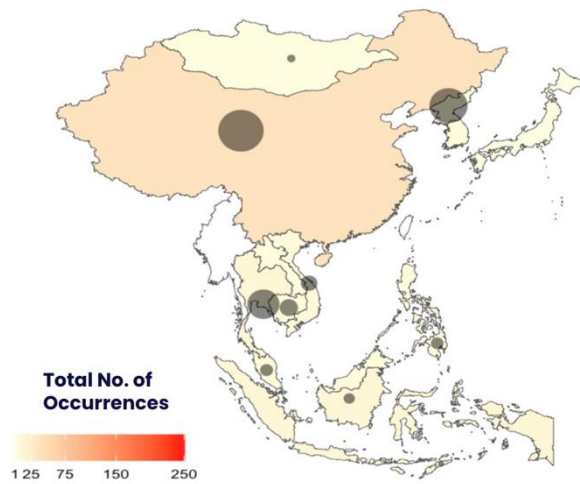
Of the natural hazards referred to using this database, droughts and wildfires affected the most people in Eastern and South-Eastern Asia, affecting 15.4 million people in China, 10.4 million people in the Democratic People's Republic of Korea and 7 million people in Thailand between 1991-2022 (Figure 6, top). These climatological events occurred between 4-38 times between these countries in the same period.

Drought and Wildfire

**Total Occurrences & Avg.
Affected Population:**
1991 – 2022
(Source: EM-DAT)

**Average No. of people
affected**

- < 100K
- 500K
- 1M
- 5M
- 10M
- >15M

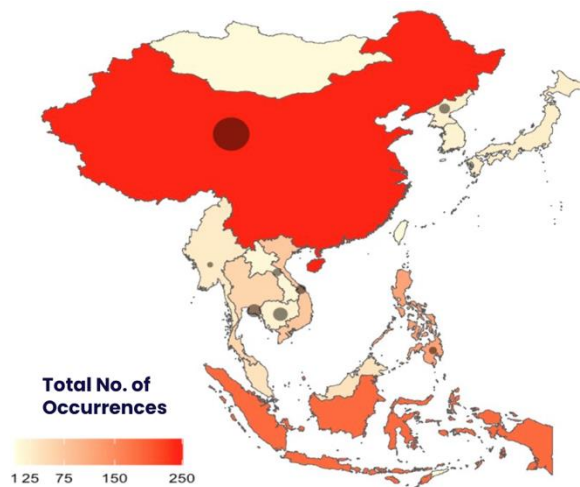


Flood and Landslide

**Total Occurrences & Avg.
Affected Population:**
1991 – 2022
(Source: EM-DAT)

**Average No. of people
affected**

- < 100K
- 500K
- 1M
- 5M
- 10M
- >15M



Storm and Extreme Temperature

**Total Occurrences & Avg.
Affected Population:**
1991 – 2022
(Source: EM-DAT)

**Average No. of people
affected**

- < 100K
- 500K
- 1M
- 5M
- 10M
- >15M

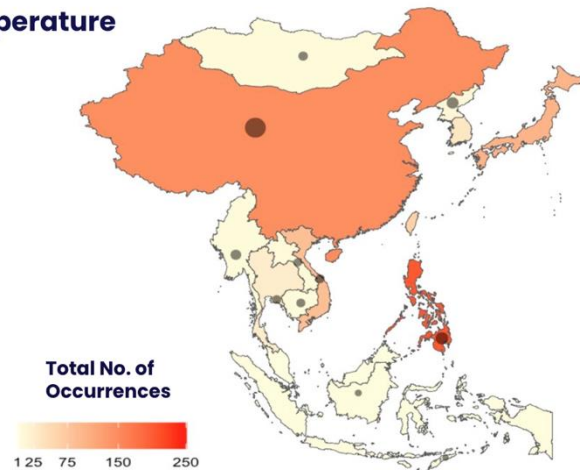


Figure 6. Number of reported droughts and wildfires (top), flood and landslides (middle) and storms and extreme temperatures (bottom) between 1991-2022. Red legend depicts the total number of occurrences of each category of hazard between 1991-2022, the black circle size is representative of the average affected population in that period.

Previous Floods

The retrospective number of floods and landslides occurring between 1991-2022 is particularly high in China, Indonesia and the Philippines as these occurred 251 times in China, 186 times in Indonesia and 119 times in the Philippines (Figure 6, middle). On average, most people were affected in China where some 9 million people experienced such effects.

Previous Extreme (Hot) Temperatures and Storms

Storms and extreme temperatures occurred most in the Philippines and China in this region (Figure 6, bottom). 2.9 million people were affected by 147 of such meteorological events in China and 8 million people were affected by 204 of these events in the Philippines.

Future Projections

Future projections are based on the middle of the road emissions scenario (SSP2-4.5 Shared Socio-economic Pathway) from the Coupled Model Intercomparison Project - Phase VI (CMIP6) multi-model GCMs ensemble provided in the IPCC, 2021. While not the worst-case emissions scenario, SSP2-4.5 assumes that the 2015 Paris agreement commitments are not achieved and 2°C global warming is not avoided. GCMs have varying climate sensitivity and generally also differ in their representation of earth system processes. While their projections in the intensity of a particular hazard in the future may therefore differ despite a persistent signal in historical observed records, the direction of the signal (i.e., a positive or negative change of a climate hazard) are generally more robust across the GCMs.

- Mean and extreme temperatures: The projected change in the number of hot days above 35°C in 2030 relative to 1986-2005 is shown to increase most by up to 40 days in Thailand, over parts of China and in Indonesia compared to the recent past (Figure 7a) (high confidence).
- Droughts and wildfires: This mean and extreme temperature projection will lengthen and intensify fire weather seasons in northern expanses of this region (low to medium confidence). There is low confidence in the intensity and frequency of hydrological droughts in large parts of East and South-Eastern Asia.
- Heavy precipitation events and flooding: Projected changes in extreme rainfall in 2030 compared to 1986-2005 show that over large parts of Asia, an increase in extreme precipitation and also in average precipitation is projected (Figure 7b) (high confidence). This is partly associated with pluvial flooding, particularly in Northern Asian Regions, and will likely increase flooding here (high confidence). Projections also show the compound effects of climate change, land subsidence, and local human activities will lead to higher flood levels and prolonged inundation in the Mekong Delta (high confidence).
- Sea level rise: Relative sea level is projected to increase associated with coastal area loss and shoreline retreat and coastal flooding (high confidence).
- Tropical cyclones: Models generally project an increase in frequency yet a decrease in intensity of Category 4-5 tropical cyclones in this region. Although the spread (indicating the range of change across models) is wide, the median increase is projected to be around 6% relative to 1986-2005 (low confidence).

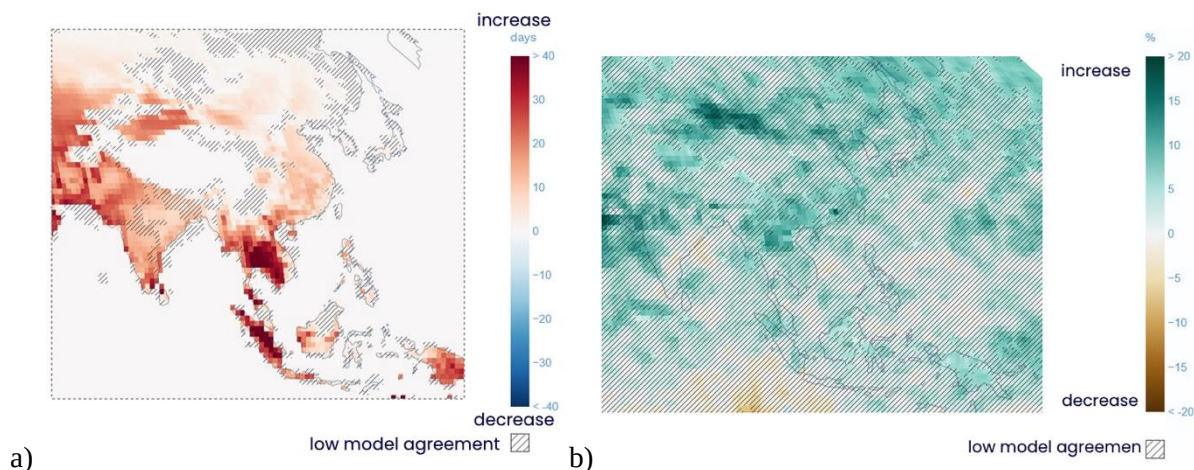


Figure 7. Projected changes in a) hot days above 35°C b) annual maximum 5-day precipitation for 2030 compared to 1986-2005 under the middle of the road emissions scenario SSP2-4.5.

Note: Detailed information on the projections and the associated confidence were accessed from the IPCC Working Group I contribution to the Sixth Assessment Report "Climate Change 2021: The Physical Science Basis" and the associated [IPCC WGI Interactive Atlas](https://www.ipcc.ch/wg1/interactive-atlas/) and regional factsheets: [Fact Sheets | Climate Change 2021: The Physical Science Basis \(ipcc.ch\)](https://www.ipcc.ch/wg1/interactive-atlas/factsheets/)

Appendix 7: Regional Needs

Top Hazards	Q1. Who?	Q2. What? (mental health and wellbeing outcomes)	Q3. How?	Q4. Climate adaptation/mitigation	Q5. Mental health actions
Extreme heat	Elderly Laborers (construction workers) People with non-communicable diseases (hypertension, DM) Farmers and fisher peoples (agriculture) School children	Heat stroke can impact physical and mental health Social security: people are more prone to psychosocial distress (getting irritated easily) Extreme heat is associated with higher rates of suicide and depression Increased temperatures	Increased exposures to heat by being outside (direct) Additional cost due to increased energy consumption Economic impacts on agricultural sector	Tree cover in urban and informal settlements Design school facilities that can adapt to heat Potable water access especially to those suffering from heat waves Climate-resilient crops Move towards more traditional and balanced architecture that	Increase communication and awareness of the link between CC and MH Green spaces to address for MH and CC Ecotherapy Community activities and spirituality as a means for recovering in disasters

	Urban poor Women	correlates to higher rates of domestic violence and irritability Anxiety Increased irritability		reduces heat with sustainable cooling materials Reforestations involving the women and the youth since we see they are mostly affected by it Rural health centers, primary healthcare - equipping with more awareness to respond to heat-related issues	Making sure political leaders are aware of this Mental health policies that are inclusive
Drought	Women (collecting water in household due to expected gender roles) Farmers and fisher peoples PWDs People with chronic medical conditions	Increased rates of domestic violence Increased PTSD and anxiety Increased rates of drowning, injuries, hypothermia, and animal bites Scarcity of food and water Agricultural occupations are negatively affected Farmer suicides from increase of debt from failed crops Patients with chronic	Nutrition (difficult to maintain healthy body and healthy mind)	Water retention practices (capture, store, generate)	Community savings , farmer insurance

		medical conditions			
Heavy precipitation, extreme weather events (e.g., typhoons), and floods	Rural Communities Urban Poor Coastal Communities Indigenous and marginalised Groups Elderly, Children, and PWDs Agriculture sector (farmers and fisher peoples) Rural residents Women People living in coastal areas, low-lying areas	Athlete's foot Communicable diseases Trauma (PTSD) Loss and grief (habitat and community loss) Economic impacts Mental health impacts Loss of homes and livelihoods Difficulty in accessing hospitals due to damaged roads and bridges Understaffed with MH professionals in Philippines, even general MH Care Displaced for evacuation - not enough MH practitioners to do MHPSS	Rural areas - access to healthcare including MH care Actions are reactive; some districts not prepared; anticipatory action is low Local govt (e.g. during COVID); allocation of money - not prioritising MH; budget distribution, which sectors are they investing more in Children - education; missing school Areas of work less accessible Infrastructure destruction due to heavy precipitation, soil erosion	Provision of community houses and evacuation centers (adaptation) Waste management to address blocking of rivers due to objects like trash (mitigation) Initiative to strengthen local health systems in the community level rather than regional Avoid community housing in riverbanks Provision of livelihood and economic opportunities Urban planning - Nature-based solution by strengthening the riparian ecosystem to mitigate floods, urban planning to capture rain water and manage proper drainage (mitigation) Structural solutions like sea walls and dykes Creating hotlines for people with	Educational interventions Trauma counseling for both responders and people affected Improve early warning systems Praying/religious/spiritual interventions Limited MH specialists - train lay-people to do counseling - has been done in other disasters Train in local language - internationally trained staff but on the ground support requires connection to those impacted by disaster and provide support on the ground Strengthen social services; if someone affected by poverty and affecting MH - need to address the poverty e.g. people finding shelter - addressing underlying issues, cascading effects

				pressing mental health needs (anxiety, suicidal ideation/attempts)	
Relative sea level rise (coastal flooding, coastal erosion)	Fishing communities Coastal communities Fisher peoples Small islands are being impacted by sea level rise such as Tuvalu.	Economic anxiety in terms of financial security Anxiety Displacement Dislocation from their communities and livelihood Climate refugees	Disconnection from their communities may result to depression and anxiety Nature-based solutions such as mangrove forests Community-led relocations/movement.	Mangroves Adaptation must be considerate of the source of livelihood especially to fisher peoples (e.g. in the relocation of communities)	Plogging builds community and at the same time it connects time to nature Community increases the wellbeing of people Inclusive and culture sensitive programs Cultural appreciation of the connection with nature Mandate the increase of MH care access in primary healthcare centers
Air pollution and water pollution	Agriculture Plantations	People stay indoors increases stress levels People with existing mental disorders will have worse symptoms		Management of agriculture waste Agroecology instead of mainstream farming	
Landslide	People living in rural areas, steep areas	People living in rural areas more likely to have houses made of light materials which are more prone to damage	Soil erosion accelerated by combined volcanic activities and typhoons, consecutive typhoon	Transfer residents to evacuation centers/safer places	Prolonged evacuation increases cases of mental health concerns; psychosocial support needed Creating hotlines for people with

					pressing mental health needs (anxiety, suicidal ideation/attempts) Increased psychological first aid training
--	--	--	--	--	--

References

- ¹ World Health Organization (WHO). *Mental health and climate change: policy brief*. (World Health Organization, 2022).
- ² Lawrance, E. L., Thompson, R., Newberry Le Vay, J., Page, L. & Jennings, N. The Impact of Climate Change on Mental Health and Emotional Wellbeing: A Narrative Review of Current Evidence, and its Implications. *International Review of Psychiatry* **34**, 443–498 (2022).
- ³ Roland, J., Kurek, N. & Nabarro, D. *Health in the climate crisis: A guide for health leaders*. (World Innovation Summit for Health, 2020).
- ⁴ Hickman, C. *et al.* Climate anxiety in children and young people and their beliefs about government responses to climate change: a global survey. *The Lancet Planetary Health* **5**, e863–e873 (2021).
- ⁵ Chen, S. X., Lee, M., McVea, D. A. & Henderson, S. B. Risk of mortality among people with schizophrenia during the 2021 heat dome. *British Columbia Medical Journal* **65**, 158–9.
- ⁶ Page, L. A., Hajat, S., Kovats, R. S. & Howard, L. M. Temperature-related deaths in people with psychosis, dementia and substance misuse. *Br J Psychiatry* **200**, 485–490 (2012).
- ⁷ Woodland, L., Ratwatte, P., Phalkey, R. & Gillingham, E. L. Investigating the Health Impacts of Climate Change among People with Pre-Existing Mental Health Problems: A Scoping Review. *IJERPH* **20**, 5563 (2023).
- ⁸ Charlson, F. *et al.* Climate Change and Mental Health: A Scoping Review. *IJERPH* **18**, 4486 (2021).
- ⁹ World Health Organization (WHO). *2021 WHO Health and Climate Change Survey Report*. (World Health Organization, 2021).
- ¹⁰ Romanello, M. *et al.* The 2022 report of the Lancet Countdown on health and climate change: health at the mercy of fossil fuels. *The Lancet* **400**, 1619–1654 (2022).
- ¹¹ Gale, N. K., Heath, G., Cameron, E., Rashid, S. & Redwood, S. Using the framework method for the analysis of qualitative data in multi-disciplinary health research. *BMC Med Res Methodol* **13**, (2013).
- ¹² Tam, K.-P., Chan, H.-W. & Clayton, S. Climate change anxiety in China, India, Japan, and the United States. *J Environ Psychol* **87**, 101991 (2023).
- ¹³ Guinto, R. R. *et al.* An agenda for climate change and mental health in the Philippines. *Lancet Planet Health* **5**, e755–e757 (2021).
- ¹⁴ Chung, S. J., Jang, S. J. & Lee, H. Eco-anxiety and environmental sustainability interest: A secondary data analysis. *Int J Mental Health Nurs* **32**, 1462–1472 (2023).
- ¹⁵ Aruta, J. J. B. R. & Guinto, R. R. Climate anxiety in the Philippines: Current situation, potential pathways, and ways forward. *JCCH* **6**, 100138 (2022).

- ¹⁶ Lally, J., Samaniego, R. M. & Tully, J. Mental health legislation in the Philippines: Philippine Mental Health Act. *BJPsych Int.* **16**, 65–67 (2019).
- ¹⁷ Waelde, L. C. *Resistance, Resilience, and Recovery from Disasters*. (Emerald Publishing Limited, 2020).
- ¹⁸ Lauber, C. & Rössler, W. Stigma towards people with mental illness in developing countries in Asia. *Int Rev Psychiatry* **19**, 157–178 (2007).
- ¹⁹ Zhang, Z. *et al.* Overview of Stigma against Psychiatric Illnesses and Advancements of Anti-Stigma Activities in Six Asian Societies. *IJERPH* **17**, 280 (2019).
- ²⁰ Ito, H., Setoya, Y. & Suzuki, Y. Lessons learned in developing community mental health care in East and South East Asia. *World Psychiatry* **11**, 186–190 (2012).
- ²¹ Intergovernmental Panel on Climate Change (IPCC). *Climate Change 2021 – The Physical Science Basis: Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. (Cambridge University Press, 2023).
- ²² Knutson, T. *et al.* Tropical Cyclones and Climate Change Assessment: Part II: Projected Response to Anthropogenic Warming. *Bulletin of the American Meteorological Society* **101**, E303–E322 (2020).
- ²³ Jimenez, M. P. *et al.* Associations between Nature Exposure and Health: A Review of the Evidence. *IJERPH* **18**, 4790 (2021).
- ²⁴ Maddock, A., Blair, C., Ean, N. & Best, P. Psychological and social interventions for mental health issues and disorders in Southeast Asia: a systematic review. *Int J Ment Health Syst* **15**, (2021).
- ²⁵ Thomas, I., Martin, A., Wicker, A. & Benoit, L. Understanding youths' concerns about climate change: a binational qualitative study of ecological burden and resilience. *Child Adolesc Psychiatry Ment Health* **16**, (2022).
- ²⁶ UN Youth. *Regional overview: Youth in Asia and the Pacific*. <https://www.un.org/esa/socdev/documents/youth/fact-sheets/youth-regional-escap.pdf> (2013).
- ²⁷ Association of Southeast Asian Nations (ASEAN). Overview. ASEAN <https://asean.org/our-communities/asean-socio-cultural-community/education-youth/> (2024).
- ²⁸ Errico, S. *The Rights of Indigenous Peoples in Asia*. https://www.ilo.org/wcmsp5/groups/public/---dgreports/---gender/documents/publication/wcms_545484.pdf (2017).
- ²⁹ Asia Indigenous Peoples' Pact (AIPP), International Work Group for Indigenous Affairs (IWGIA) & Asian Forum for Human Rights and Development (FORUM-ASIA). *ASEAN's Indigenous Peoples*. https://www.iwgia.org/images/publications/0511_ASEAN_BRIEFING_PAPER_eb.pdf (2010).
- ³⁰ Hu, X., Wu, J., Chen, P., Sun, T. & Li, D. Impact of climate variability and change on crime rates in Tangshan, China. *Sci Total Environ* **609**, 1041–1048 (2017).

³¹ Le, P. D. *et al.* Barriers and facilitators to implementation of evidence-based task-sharing mental health interventions in low- and middle-income countries: a systematic review using implementation science frameworks. *Implementation Sci* **17**, (2022).

³² CRED. EM-DAT: The International Disaster database. <https://www.emdat.be/> (2023).